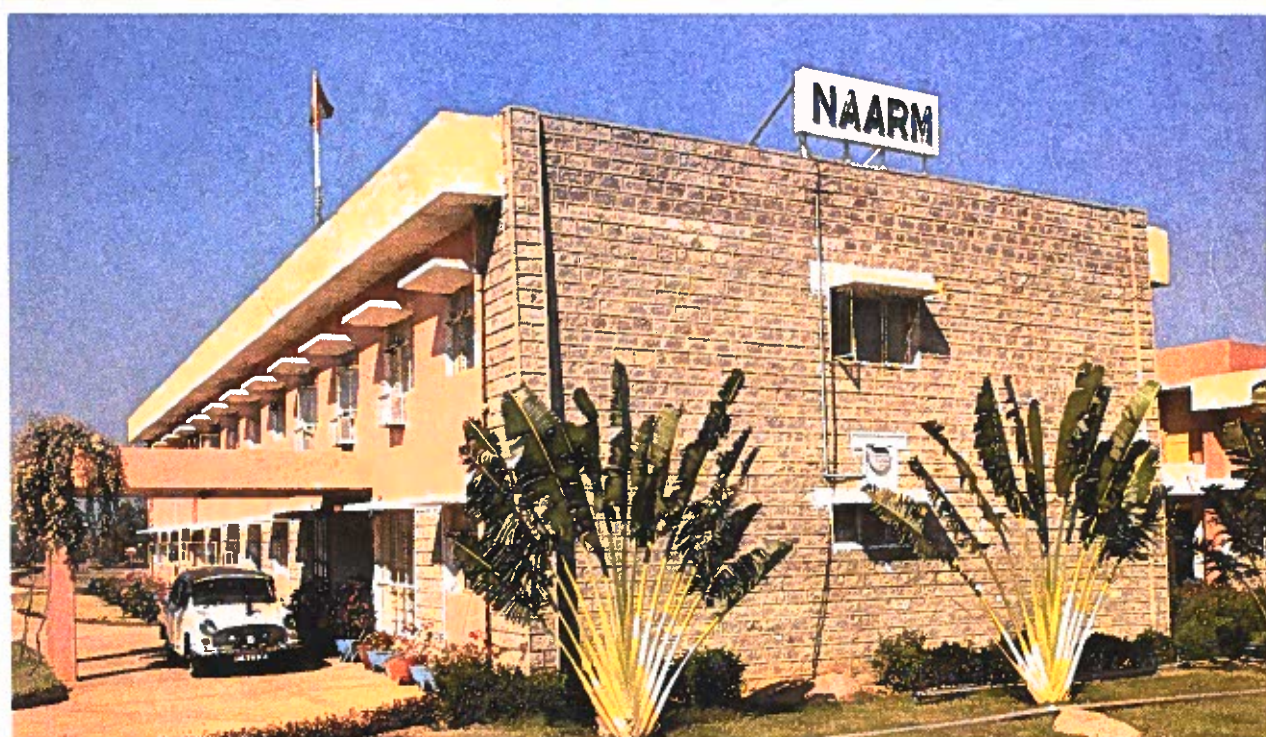
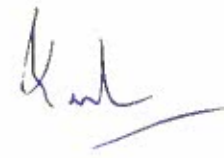


ANNUAL REPORT

1987



**NATIONAL ACADEMY OF
AGRICULTURAL RESEARCH MANAGEMENT**
Rajendranagar, Hyderabad-500 030 (A.P.)



ANNUAL REPORT 1987



**NATIONAL ACADEMY OF
AGRICULTURAL RESEARCH MANAGEMENT
Rajendranagar, Hyderabad - 500 030. (India)**

Compiled and edited by :

Dr.M.P.Singh, Professor
Sh.R.V.V.S.Prakasa Rao,
Asstt. Editor.

Cover design and photography:

Sh.A.Murali Mohan Rao,
Artist
Sh.L.Venkateswarlu,
Photographer.

Secretarial Assistance:

N.Raghunath,
Jr.Stenographer.

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1. INTRODUCTION

1.1. Historical background :

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In the present era, training in Agricultural Research Management is increasingly becoming more important as science and technology is being harnessed for augmenting agricultural production in the country. Agriculture is gradually assuming the status of an organised industry. It has, thus, become important for all those concerned with agricultural research, education and development to be familiar with the principles of scientific management. The urgent need for an institutional set up to provide such training for workers engaged in agricultural research had long been felt and the Indian Council of Agricultural Research (ICAR) and the Government of India were seized of this matter in the early 1970s. In 1969-70, the Government of India was approached by the Food & Agricultural Organisation of the United Nations to host a training course in agricultural project analysis under the regional technical assistance project of the United Nations Development Programme (UNDP) and a Regional Training Course was organised at the Indian Agricultural Research Institute, New Delhi during November 2-28, 1970. This led to the setting up of National Training Centre for Agricultural Development and Project Planning at the IARI, New Delhi in April, 1971 as a part of the Fourth Five Year Plan activity. As the scheme was a self-liquidating one, at the end of the Fourth Five Year Plan, the activities of the National Training Centre were terminated on 28th February, 1974. However, a continuation of the National Training Centre work was found desirable by the authorities.

During the Fifth Five Year Plan, the ICAR took the initiative and appointed a high powered working group under the Chairmanship of Dr. George Jacob, Chairman of University Grants Commission, to give guidelines for formulating concrete proposals for organising a Central Staff College for Agriculture to ensure continuance of the activities of the National Training Centre. The working group had representatives not only from the concerned departments of the Government of India, but also other related educational agencies. The group identified the two main objectives of the proposed Central Staff College for Agriculture as (i) to function as a repository of ideas and information, both national and international in the field of agricultural research and development and act as a clearing house for dissemination and utilisation of such information and know-how; and (ii) to undertake a programme of publishing high quality training material and allied publications as important and priority areas of action. The ICAR appointed a sub-committee of experts of the above

working group under the chairmanship of Prof. M.V. Mathur in 1975 to prepare details of the proposal. The sub-committee spelt out the need for the proposed Central Staff College for Agriculture with a fairly wide mandate and objectives.

With an outlay of Rs.50 lakhs for the Fifth Five Year Plan period, the Central Staff College for Agriculture came into existence on 1st September, 1976. Later the Staff College was renamed as National Academy of Agricultural Research Management with effect from April 1, 1979.

The Academy was started as a temporary expedient at Rajendranagar, Hyderabad where the Andhra Pradesh Agricultural University (APAU) placed at the disposal of the ICAR certain physical facilities such as hostel, class rooms and the administrative buildings. Subsequently, a broad based expert committee was appointed by the ICAR for recommending the permanent location of the Academy. The Committee visited several sites and locations and recommended Hyderabad as the permanent location since several infrastructure facilities and supporting institutions are available. For establishing the Academy, APAU released 50 hectares of land on lease for 99 years. On this land the Academy has constructed an Administrative Building, a Conference Hall, a Committee Room, an Academic Block, a 154 rooms Residence Hall, a Scientist Home and quarters for all categories of staff. The Academy is located at Rajendranagar and is 15 kms away from the centre of Hyderabad city. The campus is spread over 50 ha. of undulating picturesque landscape. The area is surrounded by APAU farms on Eastern, Northern and Southern sides and by National Institute of Rural Development on the Western side and is situated within Rajendranagar Municipal limits.

1.2. Objectives

The National Academy of Agricultural Research Management (NAARM) has been established with the main objectives of familiarising all those concerned with the Agricultural Research, Education and Development with the principles of scientific management in agricultural research and education. Since its inception it has taken up the responsibility to plan, organise and conduct training programmes, workshops and seminars in the various areas related to agricultural research management for the research scientists of the ICAR and Agricultural Universities. It has also been functioning as repository of ideas and information both at national and international level and to act as a clearing house for dissemination and utilisation of such information and know-how. The Academy has also recently

taken up the training programmes in the area of educational technology, curriculum development and structured learning. It has also taken up research programmes and case studies in various fields of agricultural research management and educational technology.

The main objectives of NAARM are as follows :

1. To organise and conduct training programmes in Agricultural Research Management to Scientists at various levels.
2. To build up high quality resource material in the subject of Agricultural Research Management based on actual field experience.
3. To undertake systematic review and study of management problems of ICAR Research Institutes; Agricultural Universities, Agricultural Research Projects and Systems.
4. To organise, liaise and coordinate programmes of international cooperation in the field of Agricultural Research Management.
5. To plan, organise and conduct training programmes, workshops and seminars in educational technology, curriculum development and structured learning for the faculty of Agricultural Universities and ICAR Institutes.

The National Academy of Agricultural Research Management functions with a wide mandate. It is a novel, vibrant and emerging institution with a visionary outlook which has a dynamic approach and pragmatic policies with realistic expectations. It functions with a zest, zealousness and commitment of purpose to fulfil the mandate of the Indian Council of Agricultural Research as the major national coordinating body for agricultural research and education, and thus fulfil the national aspiration for better management of science and agriculture.

2. TRAINING PROGRAMMES

The Academy has been planning, organising and conducting various types of training programmes to meet the need of the clientele groups working at different levels of management in the agricultural research and development organisations. Mainly three levels of clientele have been notified for training purposes, Junior, Middle and Senior.

2.1 Courses for Junior Level Agricultural Scientists and Development Workers

Two types of courses have been designed and are being offered. These two courses are :

2.1.1. Foundation Course on Agricultural Research Management (FCARM)

The Foundation Course for the scientists recruited to the Agricultural Research Service by the ICAR is one of the regular course offered by the Academy. This Course aims at providing them exposure to concepts and principles of project management with special emphasis on project formulation and implementation as well as instruction in related areas such as group dynamics, interpersonal relations, interdisciplinary project formulation, motivation management and such other topics as would make them a more effective research workers. The Course is organised within the frame work of the following objectives.

- To acquaint the scientists with the general Agricultural Scenario in the country with reference to Research and Development.
- To present an insight to the rural society and structure.
- To equip them with the techniques in Research Project Management.
- To provide skills in organisational behaviour for high motivation, team work, conflict resolution etc.
- To provide training in agricultural communication and transfer of technology.

- To orient the scientists to the Indian Council of Agricultural Research and Agricultural Research Service.
- To foster a feeling of fraternity among the members of the ARS serving in different parts of the country.

The total duration of the course is five months. The trainees undergo six weeks training (First Phase) at the Academy oriented to the basics of Agricultural Research Management. After this training they spend about two months (Second Phase) at a Regional Research Station of National Agricultural Research Projects or in ICAR Institutes where they receive practical training in project identification and formulation and develop an appreciation for rural life and environment. They return to the Academy for a further programme of six weeks (Third Phase) where they get exposure to areas which will help them to become better research workers. During this training programme the trainees are exposed mainly to the subjects like Agricultural Research Management; Project Management, Organisational behaviour, Communication, Transfer of Technology, Administrative Rules and Procedures and Practical Classes on Audio-Visual Aids, Photography, Research Project Formulation, Indexing etc.

Agricultural Research Service (ARS) Scientists recruited through ASRB examination numbering 250 scientists (XXI to XXVIII/FCARM batch) were given intensive training in various aspects of Research Project Management viz. Research Programme Planning, Appraisal, Programme Evaluation and Review Technique, Critical Path Method, Conflict Management etc. Each participant was given training in the preparation and presentation of research projects on the basis of the information collected and training received during the Field Experience Training.

Course contents

The following Course contents were covered during the training.

I. General Agricultural Scenario in the country

- NAARM and its role
- Agro-ecological zones and cropping systems in the country
- Economics of Indian Agriculture

- Agricultural Prices
- Land Systems and measures of reform
- Agricultural Production - potential and constraints

II. Rural Society and Structure

- Rural development - philosophy and programme
- Characteristics of Indian rural life in comparison to urban life
- Rural leadership and rural development programmes
- Socio-economic surveys
- Tools of data collection for rural socio-economic surveys
- Preparation of viable research projects based on socio-economic surveys.

III. Research Project

- Agricultural Research Management Process
- Creativity and Problem solving
- Research Programme Planning
- The Research Project
- Project Appraisal
- Project Selection Exercise
- Project Management Techniques - CPM & PERT
- Resource Smoothing
- Critical Path Index
- Uncertainties in Research
- Management of Research Stations

IV. Development, Transfer and Adoption of Technology

- Technology development process - promoters and retarders
- Scientists and Transfer of Technology
- Institutional Structures for Transfer of Technology
- Treatment of Research Findings
- Semantics in Scientific Communication
- Communication through Audio-Visual Aids
- Agricultural communication through A.I.R.
- Agricultural communication through Doordarshan.

V. History, Structure and Organisation of Agricultural Research

- History and growth of ICAR
- Structure and organisation of ICAR
- Agricultural Research Service
- National Agricultural Research Project
- Linkages and coordination between ICAR and Agricultural Universities. Role of ICAR
- Linkages of ICAR with other National Research Organisations
- Research Systems in Agricultural Universities
- Farming Systems Research
- All India Coordinated Research Projects of ICAR
- Constraint Analysis in Research and Adoption.

VI. Team Work and Interpersonal Relationships

- Individual and the organisation
- Groups in Organisations

- Team Work
- Group Dynamics
- Motivation and Behaviour
- Motivation and Work
- Transactional Analysis
- Conflicts in Organisation
- Fatalism or positive thinking
- Case Analysis for Problem solving and decision making.

VII. Training in peripheral areas

- Communication Skills
- Rapid reading and comprehension
- Writing of Scientific Paper
- Basics of computer programmes and use of computer in Agricultural Research
- Programming in BASIC and FORTRAN
- Energy Management in Agriculture
- Solar and Wind Energy
- Biomass and Energy
- Pollution
- Basic concepts of Linear Programming

VIII. Transfer of Technology

- Adoption of technology - stimulants and constraints for adoption
- Development communication in Agriculture

Transfer of Technology Experience of Developed and Developing countries

- Rural Development Programme in India.

IX. Organisation of Research

- Agricultural Research Systems in the World
- International Research Organisations
- Research Planning and Coordination in ICAR
- Contribution of ICAR to Agricultural Education Systems in the country
- ICAR and International Collaboration in research.

X. Information and Library

- Source of Agricultural Information
- Citation Indexing
- Management of personal documents
- Book reviewing in the sciences
- Review literature
- Proof reading
- Poster sessions
- Technical Report Writing

XI. Trends in Research Work on some important crops/areas

- Crop Sciences
- Oilseeds Research
- Dryland Agriculture
- Pest Management
- Soil Management
- Fisheries Management
- Animal Management

- Horticultural Crops.

XII. Administration and Accounts

- General Service conditions - An Overview
- Stores - Management and Purchase Procedures
- Project budgeting and monitoring.

2.1.2. Orientation Course in Agricultural Research Management (ARMOC)

An Orientation Course in Agricultural Research Management (ARMOC) of one month duration is being conducted for the scientists working in various institutes of the ICAR. The objectives of the Course are as follows :

1. To discuss and acquaint with the issues relating to agricultural research management process.
2. To equip and acquaint with the techniques to do the work better as researcher.
3. To instil the philosophy of living, thinking and working together and foster a feeling of fraternity among the members of the ARS.

The topics in this Course broadly cover the Management Process, Research Project Management; Quantitative Techniques; Organisational behaviour, Management of Information System and Transfer of Technology.

Two orientation courses on Agricultural Research Management for 94 S-1 inductees and promotees from S to S-1 were conducted during the year. The participants were exposed to various techniques of Agricultural Research Management.

Course Content

The following course contents were covered during the training :

- Agricultural Research Management
- Project Management
- Organisational Behaviour

- Communication
- Transfer of Technology

2.2. Courses for Middle Level and Senior Level Scientists

2.2.1. Course on Agricultural Research Management (CARM)

It has been realised that the fulfilment of the national need for increased agricultural production requires that the Agricultural Scientists should organise and orient their efforts to rapid achievements of national goals, to utilise problem-centred rather than discipline-centred approaches to agricultural research and to take the initiative in arranging cooperation with all other agencies concerned with the agricultural development. Agricultural Research should be seen in developmental frame work including not only basic and applied investigations but the entire process of development of appropriate technologies and their wide spread dissemination.

The Academy is organising this Course on Agricultural Research Management (CARM) to increase effectiveness of each participant in his present position and prepare him for greater managerial responsibility by apprising him of the various aspects of research project management techniques and behavioural skills. The Course is based on the linkages between two groups of inter-related functions i.e. the functions of the agricultural research institutions and functions of a research manager.

2.2.2. Course on Agricultural Research Project Management (CARPM)

The importance of Agricultural Research in contributing to increased productivity is being fast recognised. Self-reliance and self-sufficiency are the main thrusts in agricultural production. Research and Development is to be geared up to provide technical service to the production units and help the farming community to produce more with economically feasible inputs. The research project managers have to formulate programmes, plan the projects, implement them, monitor and revise them systematically and scientifically. Appropriate linkages and communication channels with other units/divisions within the outside the research organisation to facilitate transfer of appropriate technology to the clients/farming community are necessary. An understanding of the above factors helps the research project leaders to develop a pragmatic approach to problems and make them more effective members of research team.

The Academy is organising this course on Agricultural Research Project Management (CARPM) to expose the research project managers in the ICAR Institutes to the concepts of research projects and application of the 'management approach' to the problems of agricultural research units. The course is designed to serve the interests of senior research managers heading a division in the ICAR Institutes in the cadre of S3/S4 and the duration of the course is about two weeks.

2.2.3. Course on Human Resources Management

Agricultural Research has played a key role in increasing the food production threefold from land, water and other sources in India in the past two decades which resulted in Green, White and Blue revolutions. India is not only self-sufficient in foodgrains now but is in a position to export agricultural products to other countries. India has the largest manpower of Agricultural Scientists in the world about 6480 in the ICAR system and 24,000 in the Agricultural Universities supported by an equally large technical staff. For optimum output this group has to be properly motivated and their work is to be coordinated for achieving the objectives. This involves the management of human resources (scientists, technical and supporting staff) besides financial resources. Day by day Agricultural Research is becoming more specialised, complex and sophisticated. So planning and development of human resources for this purpose is very important.

The Academy is organising the Course on Human Resources Management to expose the Research Managers to the concept of human behaviour in organisations (Organisational Behaviour) and attempt to address the issues for human resources management in Agricultural Research. The course is designed to serve the interests of the Research Managers and Senior Scientists heading a division in the ICAR Institutes in the cadres of S-3/S-4 and the duration of the course is two weeks.

2.2.4. Course on Instructional Technology and Teaching Methods

Teachers today are facing the challenges on the modes of designing instructional systems by which conditions could be deliberately created to enable the learners achieve the desired educational goals in a specified manner. Teachers are Educational leaders and Decision Makers influence the students as well as subject matter. In addition to having the command on subject matter, teachers should also be familiar with the broad spectrum of instructional methodology, techniques and approaches that can be easily applied in the classroom. The teacher should be able to analyse the educational

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training needs, keeping in view, the predetermined goals and working out instructional objectives, developing lesson plans, designing instructional material and teaching modes to be used and finalisation of suitable monitoring and evaluation techniques.

The Academy has recently started this Course with a view to expose the teachers to the need for appropriate planning of instructional strategies and optimum use of instructional resources keeping in view the complexities of teaching learning process coupled with the increase in number of students, the various educational streams from where they came as well as ever expanding horizons of modern knowledge.

This course on Instructional Technology and Teaching Methods is conducted for the senior faculty members of Agricultural Universities/ICAR Institutes who have interest in instructional technology and production of instructional aids. The duration of the course is two weeks.

2.3 Summer Institutes

The Academy organises Summer Institutes in the field of Agricultural Research Management, Education and Instructional Technology for the benefit of the Faculty Members of the Agricultural Universities and ICAR Institutes. The Summer Institutes are usually of three weeks duration.

2.4 Workshops and Seminars

The different units of the Academy are planning, organising and conducting various types of workshops and seminars related to specialised topics in their own area of specialisation. The workshops and seminars are conducted on the basis of the research studies undertaken by them. The case materials are also developed in advance for the benefit of the participants of workshops and seminars.

2.5 Consultancy Services

The Academy has been providing consultancy service in the area of Agricultural Research Management and Educational Technology. The Academy has been helping Agricultural Universities and ICAR Institutes from time to time in conducting training programmes in these areas. The Academy's Faculty are specialised in various sub-disciplines and deliver special lectures in the Institutes who arrange training programmes to their own staff. Institutes have found these lectures useful and periodically call on the services of the Faculty for this purpose.

3. STRUCTURAL ORGANISATION AND SALIENT ACTIVITIES

3.1. For the purpose of imparting effective training and undertake research studies in a meaningful way, the Academy is organised into six subject matter oriented functional units with a strong group of centralised services, as under :

3.1.1. Agricultural Research Systems & Policies

3.1.2. Educational Systems and Technology

3.1.3. Transfer of Technology Systems & Policies

3.1.4. Project Management

3.1.5. Human Resource Development

3.1.6. Information and Documentation

3.1.7. Centralised Services

(i) Farm Services and Agro-Technology Blocks

(ii) Production and Publications

(iii) Administration and Accounts.

All the above mentioned units have been conducting training programmes for Agricultural Service Personnel, Agricultural Scientists from Research Institutes and Agricultural Universities, Administrators and Technology Transfer Personnel. The Organisational Chart of NAARM is given in Fig. 1.

3.1.1. AGRICULTURAL RESEARCH SYSTEMS & POLICIES UNIT

Background/Introduction

Research system is the backbone for agricultural development. It is the technology generated and information gathered and principles and procedures evolved by research system that are utilised by the other agricultural systems like education, training, extension, client and support. How the various research systems function and their effectiveness of functioning are of utmost importance. The Agricultural Research Systems & Policies Unit studies the various systems of agricultural research in order to strengthen the various research systems that are essential for agricultural development.

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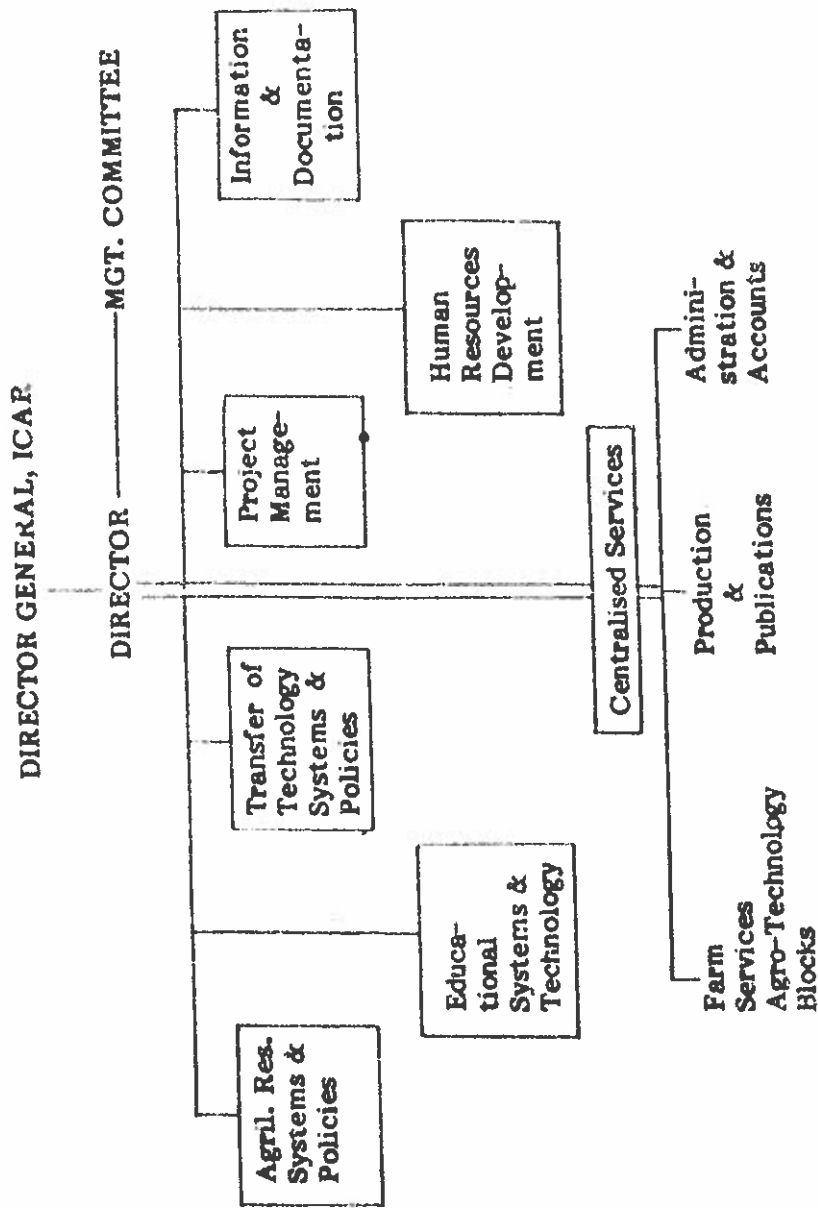
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ORGANISATIONAL STRUCTURE

Fig.1:



Objectives, Tasks and Functions

1. To conduct systematic research in the areas of Agricultural Research Systems and Policies;
2. To organise specialised training programme in systems approach with regard to specific problems of Agricultural Research Management;
3. To evolve a package of empirically proven, reliable and valid research policies to be recommended for implementation by the concerned authorities;
4. To offer consultancy advisory services to the various research agencies for effective functioning of various research systems in areas like manpower planning and utilisation etc.

Activities during the year 1987

1. Comparative study on the Agricultural Research Systems of the World :

An in-depth analysis of various Agricultural Research Systems in the world including India has carried out on the basis of information collected from various secondary sources. The final report is under preparation.

(Dr. T. Balaguru)

Study on Scientific Manpower Planning

The Unit has taken up a Project on Scientific Manpower Planning in ICAR Research Institutes. The specific objectives of the study are :

- i) To trace the growth of man power resource in different disciplines in selected research institutes of the ICAR system
- ii) To find out the relation between the objectives of the institutes and the available manpower in selected institutes
- iii) To assess the degree of utilisation of present manpower
- iv) To estimate the ratio between manpower availability and achievement of objectives

- v) To workout the time lag between manpower planning and its implementation and the reason for the lag and its impact
- vi) To predict the manpower requirement for the coming decade for selected institutes
- vii) To elicit suggestions for effective utilisation of present manpower

The data were collected from different institutes. Coding and analysis of the data is in progress.

(Dr.VE. Sabarathnam)

Measurement of Perception of Scientist

Two tests were devised to measure the perception of the Foundation Course Trainees of XXV, XXVI & XXVII batches regarding the technology development and treatment of research findings. The analysis of the data is in progress.

(Dr.VE. Sabarathnam)

Course Material Developed

The following course materials were developed and delivered by the unit to various courses conducted during the year 1987.

1. Role of Technical Cells in the treatment of research findings
2. Manpower planning for Agricultural Research
3. Utilisation of rural leadership in development programmes.

3.1.2. EDUCATIONAL SYSTEMS AND TECHNOLOGY UNIT

Background/Introduction

In recent years India has made rapid strides in imparting higher agricultural education through a network of 25 Agricultural Universities, about 100 Agricultural Colleges and ICAR Institutes. Every year a large number of agricultural graduates and postgraduates are coming out of these Institutions. Though the Agricultural Universities have inherited and adopted the educational pattern of Land Grant University

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of U.S.A. i.e. the integration of teaching, research and extension education functions, it is felt that the instructional methods adopted in various Universities differ in their content and methods of imparting education and training. There is an urgent need to make higher agricultural education more meaningful and purposeful. It is realised that though the teachers are academically qualified, many of them follow the traditional methods of teaching which do not meet the demands of practical utilisation of knowledge gathered.

Currently the country is facing new challenges in the agricultural front owing to the sudden explosion in scientific knowledge and population and, in turn, to tune up the curriculum of agricultural education suitably. Hence, there is a need to study the issues relating to the content, the structure and the function of the Agricultural Curriculum at the undergraduate and postgraduate levels of education offered in our Agricultural Universities and to explore the ways of making it more purposeful and responsive to the needs of changing time.

The primary function of this Unit is to render assistance to Agricultural Universities and ICAR Institutes dealing with education, support for development of better teaching techniques. Modern education technology interacts closely with subject matter specialists to develop viable and useful curricula courses and teaching techniques.

Objectives, Tasks and Functions

1. To conduct training in Education Technology as applicable to curriculum
2. Educational planning, administration and management
3. Instruction techniques and material production
4. Studies on teaching-learning process including evaluation
5. Consultancy and Advisory services

Activities during the year 1987

The following course was planned, organised and coordinated in the area of Instructional Technology.

Course on Instructional Technology and Teaching Methods

A two weeks Course on Instructional Technology and

Teaching Methods for senior faculty members of Agricultural Universities was organised from February 10-24, 1987. The course was designed to expose the teachers to the need for appropriate planning of instructional strategies and optimum use of instructional resources keeping in view the complexities of teaching learning processes coupled with the increase in number of students, the various educational systems from where they come as well as ever expanding horizons of modern knowledge.

Twenty senior faculty members of Agricultural Universities, interested in instructional technology and production of instructional aids participated in the course.

Course material developed

1. Principles of effective class room instruction
2. Curriculum Development & Design
3. Conceptual frame work and guidelines for curriculum development
4. Role of Audio-Visual Aids in Teaching
5. Preparation, planning and use of Audio-Visual aids
6. Instructional methods
7. Problem solving in teaching-learning process
8. Innovative teaching methods

Consultancy and Assistance Provided to Agricultural Universities and ICAR Institutes

1. Consultancy provided to Assam Agricultural University Jorhat in establishing a Television studio for instructional purposes.
2. Assisted Orissa University of Agriculture and Technology in developing suitable curricula related to agricultural research management courses.
3. Consultancy provided to the Extension Education Institute, Andhra Pradesh Agricultural University, Rajendranagar, Hyderabad in conducting indepth study of impact of training programmes conducted at the Institute.

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5.1.3. Transfer of Technology Systems and Policies Unit

Background/Introduction

The Primary function of this unit is to impart training and conduct research on Technology Transfer System. The case studies on existing methods of transfer of technology such as Operational Research Project, Lab to Land, KVK System are taken up by the Unit.

Objectives, Tasks and Functions

1. Training in Technology Transfer Processes and Appropriate Technology selection
2. Develop effective Agricultural Communication techniques and mechanism
3. Identify bottlenecks and managerial deficiencies of TOT Systems and evolving more efficient processes.
4. Consultancy and Advisory Services.

Activities during the year 1987

Several studies on (1) Management of Lab to Land Programmes of ICAR; (2) Training needs of ICAR Scientists; and (3) Communication and job satisfaction prevailed in ICAR Institutes have been taken up by the Unit. Some results of the same have been published and some are under process of finalisation for publication.

(Dr.R.K. Samanta)

5.1.4. PROJECT MANAGEMENT UNIT

Background/Introduction

Project management is designed to integrate complex efforts and deals with planning, scheduling, directing and controlling of institute resources for a project that has been established for the completion of specific goals and objectives. The project management utilizes the 'system approach' to management though the use of personnel belonging to different functional disciplines assigned to a specific project. The main objective of project management is to make the most efficient and effective use of the resources of manpower, equipment, facilities, materials, money, information/

technology so that the Institute objectives and goals can be achieved as per schedules, budget and the desired performance, taking into consideration the ever changing environmental input factor.

The integrated planning and control systems assure that the functional disciplines will understand their total responsibility towards achieving project needs; assure that problems resulting from scheduling and allocation of critical resources are known before hand. The problems that can jeopardise successful project completion can be forecast, so that effective corrective action can be taken to prevent or resolve the problems.

The Project Management unit provides an individual research worker the necessary tools and techniques by which he can find proven, workable solutions to problem that can occur in a project management.

Objectives, Tasks and Functions

The objectives of Project Management Unit are :

1. Training on Agricultural Research Project Management Techniques
2. Case studies on planning, monitoring, controlling and evaluation of Agricultural Research Projects.
3. Network Analysis - Flow Chart and scheduling of the project activities and the networks of the projects
4. Project budgeting and financial control
5. Consultancy and advisory services.

Activities during the year 1987

1. Course on Agricultural Research Project Management (CARPM)

Three courses on Agricultural Research Project Management were organised from Jan. 16-30, 1987; March 2-13, 1987 and August 8-23, 1987 for the benefit of the 82 senior scientists of ICAR in the cadre of S-3 and S-4 and the Professors and Directors of Research, Agricultural Universities.

COURSE CONTENT

The following course contents were covered:

- Project Planning - Importance of planning long-term and short-term project in the management process and systematic project planning
- Project selection - Methods and criteria used for project selection so that one can undertake better research projects in agriculture
- Monitoring and Evaluation - Network methods for time scheduling; control and monitoring; and resource smoothening of Research Projects.
- Organisation Structure - Principles of organisation and the importance of technology; and environment in organisational design.
- Group Dynamics - Characteristics and classification of groups; group decision making; and inter-group problems in organisations.
- Leadership - Role of leadership in management; leadership styles suitable to Indian conditions.
- Conflicts - Types of conflict that occur in organisations; and the methods used to resolve the conflicts.
- Organisational Communication - Meaning, concept and importance of organisational structure; communication behaviour; organisation structure's effect on communication; avoiding resentment; principles of effective listening; and barriers to listening.
- Mass Communication - Television in rural communication; steps in developing a TV programme; and rural broadcasting.

WORKSHOP

WORKSHOP ON NATIONAL AGRICULTURAL RESEARCH PROJECTS

A Workshop on national Agricultural Research Projects (NARP) was organised from February 24-26, 1987. The objective of the workshop was to strengthen permanently the capabilities of State Agricultural Universities to conduct location speci-

fic, production oriented research in the agro-climatic zones identified in their service areas. It would achieve this through a reform of existing university - based agricultural research and would concentrate in particular on research oriented towards the needs of the local agro - ecological zones with special emphasis on rainfed food grains and oilseeds. Some case studies on performance and evaluation of NARP sub-project were presented at the Workshop.

Vice-Chancellors of Agricultural Universities, Directors of Research and Extension all over the country numbering thirty six participated in the workshop.

BOOK ON 'PROJECT MANAGEMENT TECHNIQUES'

A book on 'Project Management Techniques' in R & C for Agriculture, Veterinary and Fisheries is being written. The following topics have been completed:

Part I

Gantt Bar Charts; Activity-Slack Bar Chart; Activity Bar Chart; Terminology of Networks; Rules for Preparation of Networks; Preparation of Networks and their uses; Network Analysis; Critical Path; How PERT and CPM Differ; PERT Concept and Application; Critical Path Index; Critical Path Method Precedence Diagram; Floats; Job Boundary limits; Resource Smoothing, PMTs in ICAR and SAUs.

Part II

A case for illustration of ASBC; Cases for illustration of PERT; Cases for illustration of CPM; A case for illustration of CPI; A case for illustration of PD; Cases for illustration of Job Boundary limits.

2. STUDY ON FEASIBILITY OF INTRODUCING PROJECT-WISE BUDGETING IN ICAR INSTITUTES

Study on feasibility of introducing project-wise budgeting in ICAR institutes was undertaken. The work was undertaken with the help of the private consultants M/s. Ferguson & Co. The first phase of the study was taken up to understand the existing systems and for suggesting the suitable modifications

(Dr.P. Manikandan

5.1.5. HUMAN RESOURCES DEVELOPMENT UNIT

BACKGROUND/INTRODUCTION

People form the most important resource in any organisation. Human Resources Development unit is concerned with methods and approaches for the growth and development of the people in organisations towards higher level of competency, creativity and fulfilment. The concept of development covers not only individuals but also groups. The Human resources approach helps people grow in self control, responsibility and other abilities and then creates a climate in which all employees may contribute their best for improved results and efficiency. This is applicable to research organisations and scientists. Human resources development is attempted through training, feedback and counselling. Human resources are under-utilised in many organisations.

OBJECTIVES, TASKS AND FUNCTIONS

1. To improve the research productivity and efficiency of Agricultural Scientists and Teachers through behavioural changes by training in human resources development and organisational behaviour. They are exposed to following topics:
 - Motivation
 - Interpersonal relationships team work
 - Leadership
 - Conflict Management etc.
2. To conduct research in the areas of Human Resources Management and Organisational Behaviour relevant to Agricultural Research and Education Management.
3. To prepare case studies, structured experiences for training in human behaviour, audio-visual aids and Role play for effective training.
4. To provide consultancy in the areas of Human Resources Management and Organisational Behaviour.

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Dr.P. Manikandan)

ACTIVITIES DURING THE YEAR 1987

Course on Human Resources Management

Two courses on Human Resources Management were organised from April 8-23, 1987 and October 5-17, 1987 to expose the Agricultural Research Managers to the concepts of human behaviour in organisations and attempt to address the issues for human resources management in Agricultural Research.

The courses dealt with the Management of Human Resources and covered the following topics :

Individual and the Organisation; Motivation and scientists; Interpersonal Relationships; Transactional Analysis; Group Dynamics; Leadership; Team Work; Conflicts; Personnel Policies; Structure and Design of Organisation; Organisational Communication; Mass Communication; Research Project Management etc.

Fifty four senior scientists and Research Managers heading a Division in the ICAR Institutes participated in the courses.

3.1.6. INFORMATION AND DOCUMENTATION UNIT

BACKGROUND/INTRODUCTION

Information and documentation from the most critical inputs in any form of management activity. Such organised scientific information base is very important for decision making at all levels. Besides, a personal scientific information and retrieval system is necessary for every scientific worker to function as a more effective researcher.

OBJECTIVES, TASKS AND FUNCTIONS

1. To function as a repository of ideas and information, both national and international, in the field of Agricultural Research Management and Education and develop as a clearing house for dissemination and utilisation of such information to the clientele institutions.
2. Training Scientists in the personal scientific information management including the user education users programmes.
3. To develop appropriate management information systems for scientific management.

4. To coordinate for the sectoral information centres in India in the field of Agricultural Sciences.
5. To bring out learner's information packages and transferring Workshop/Training kits.
6. Agricultural Research Management Abstracts.
7. Consultancy and Advisory Services.

ACTIVITIES DURING THE YEAR 1987

Technical Services

At present the Unit has about 17,000 publications and receiving 400 Journals on subscription/Exchange basis. The Unit has many unique features in its information storage and retrieval. The following specialised corners have been formed for the effective utilisation of the documents.

Foreign Tour Reports Corner

In order to have an access for the Information on Agricultural Systems Studies of the different countries, a specialised corner for Foreign Tour Reports of the ICAR/DARE Scientists has been established. A copy each of the tour reports after submission to ICAR is being sent to NAARM and these are organised countrywise with additional entries on card form by Scientist's name, Institutes and areawise. It is planned to introduce reprographic services for such information and also to microfilm these reports for effective storage and utilisation.

International Corner

Efforts have been made to establish repository centres of all institutes coming under CGIAR group as also many international organisations engaged in Agricultural Research, Development, Training, Education etc. This corner has preserved all the important publications of the International Agricultural Research Institutes for the benefit of all engaged in Agricultural Research and Development. Periodically catalogues of these holdings will be made available for effective utilisation.

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Reprographic Section

This section of the Unit caters to the reprographic needs of the entire Academy and the users with four sophisticated plain paper copiers having enlargement and reduction facilities, three electronic stencil cutters and one electronic adhesive binder. About 1,70,500 copies were reproduced from the plain paper copiers and 1,500 electronic scanning stencil were cut for the use in training programmes.

Newspaper Clipping Service

Under this service, items of Agricultural Education, Research Management, Technology, Project Management, Information Management, Human Resource Development etc. are cut and displayed. The displayed clippings are filed and kept for future reference on areawise.

Training

Scientific information management is an integral part in almost all the training programmes organised at the Academy. The Unit has developed many interesting topics for the benefit of scientists in the ICAR Institutes and the Faculty Members of the Agricultural Universities. Different programmes were developed based on the levels of participating scientists. The sessions dealt were problem oriented and emphasised practical use.

Agricultural Research Management Abstracts (ARMA)

This publication is for the benefit of those engaged in Agricultural Research Management, Administration, Education and Development. ARMA covers many interesting areas in Agricultural Research Management covered under the different units of the Academy.

The material for two issues of 1987 has been released and sent for publication. The ARMA is brought out by Information and Documentation Unit of the Academy and is compiled and edited by Sri R.B. Gaddagimath, Librarian of the Academy. About 700 copies are made and circulated on exchange basis.

WORKSHOPS

Workshop on Information Services Management in ICAR Institute Libraries and Information Centres

A six day workshop on 'Management Techniques' for Information handling in technical cells/sections of ICAR Institutes"

was organised from September 14-19, 1987. Shir R.B. Gaddagimath coordinated the workshop. The resource persons for the workshop were mainly drawn from the faculty, of the Academy. In addition, some of the leading specialists were invited as guest speakers.

The Workshop highlighted the improvement of the information services for the scientists and modernizing the centres to meet the changing demand and needs. During the workshop the participants discussed the systematic organisation of literature, technological developments in information handling techniques including computer application for storage and retrieval to provide effective information services to scientists. The following topics were covered during the workshop:

Information needs of the Agricultural Scientists; Management of Agricultural Scientists; Management Information Systems in Agriculture; Planning and Organisation of Inhouse documentation services; Educational Training for Agricultural Information handling and project management techniques, scientific editing and reporting, conference planning etc.

Consultancy and Advisory Services

The consultancy and Advisory Service was rendered by the Unit to a number of Institutions at the National level.

1.1.7. CENTRALISED SERVICES

Unit of Farm Operations and Management and the Agro-Technology Demonstration Blocks

Introduction

The Academy is landscaping its campus as a model for research and Educational Institutions. This is done, conscious of the fact that a beautifully landscaped campus leads to a comfortable environment and provides for improvement in micro-climate. The psychological comfort of green and sylvan surroundings promotes an academic atmosphere and leads to higher productivity. In landscaping the campus, stress is being laid in utilizing a large varied collection of trees, shrubs, climbers, herbaceous plants and ground covers with an eye on the educational value of a well labelled rich collection of flora.

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ICAR Institute

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The Agro-technology Demonstration blocks of NAARM have been conceived as the show window of the National Agricultural Research System. The latest technology and crop varieties being released by the different ICAR Institutes and Agricultural Universities are being put on show here in one place (subject to Agro-climatic limitations) for the benefit of visitors and trainees of NAARM consisting of National and International Scientists, Policy Makers and Administrators of Agricultural Research and Development.

Objectives, Tasks and Functions

The Farm Operations and Management is a centralised service of the Academy and has the objectives of landscaping of the campus, developing and maintaining the estate and laying out the Agro-Technology demonstration blocks in different areas of agriculture which includes Field crops, Fruit crops, Vegetable crops, Tuber crops, and different Sylvi systems. The land scaping activities cover the entire 50 ha. of the campus and the agro-technology demonstrations covering an area of 15 ha. of land.

Activities during the year 1987

Land Scaping :

The avenues on the major roads which have been planted are being regularly maintained. These include tree species of about 20 genera. The landscaping in and around administration block has been completed. Work is in progress in different places particularly Academic Block, Halls of Residence, Scientists Home and newly coming up Faculty Building. A rose garden of over a thousand bushes have been well established. Work is in progress to establish another rose garden with all India varieties.

Field Crops :

These demonstrations include oilseeds, millets, cereals, pulses. During Rabi 1987 demonstration of maize Var : Ashvani and Amber pop-corn); Jowar (Var ; Spv - 462, CSH - 9) Sun flower (Var : Morden, Surya, E.C. 68414 and M.S.F. 1 - 8) were taken up. Because of the prevailing drought conditions cultivation was limited for the above crops only.

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Horticultural Crops

[I] Fruit Crops :

The fruit orchards of six different fruit crops have been established. All these improved varieties are released by ICAR Institutes/Agricultural Universities. The **Guave** block is planted with the varieties of Allahabad Safeda, Lucknow 49, Safaid Jam, Kohir Safeda, and Banarasi Surkha.

The **Sapota** plot has been established with the varieities of Oval, Cricket Ball, Bombay and Kalipatti.

The **Grape** orchard has been planted with Gulabi, Cardinal, Champion, Beauty seedless, Perlette, Delight, Thomson seedless, Pusa seedless, Himrod, Banqui-abad, Early muscat, Rabinia, Kishmis Rosovis, Golden Queen, Arka Mans, Arka Shyam, Arkavati, Arka Kanchan, Dilkhus, and Anab-e-shahi.

The **Ber** plot has been planted with the varieities Gola, B.H, Mundia, Kaithaly, Umran.

The **Pomegranate** varieties grown are Jodhpur Red Basin seedless, Jyoti (G.K.V.K. - 1) and Ganesh.

The **Citrus** cultivars planted include Moosambi, Santara, Nagazilime, Khasi Mandarin, Sathgudi etc.

[II] Vegetable Crops

Forty vegetable varieties of eighteen kinds have been demonstrated during Kharif and Rabi 1987. The different kinds of vegetables grown were tomato, brinjal, chillies, bhendi, cucumber, bottle gourd, bitter gourd, sponge gourd, ridge gourd, ash gourd, cabbage, cauliflower, palak, methi, cluster beans, beans, cowpea and pumpkings.

[III] Tuber Crops

The demonstrations include **Tapioca** varieties S-1310, H-2164, H-97, H-2304, H-226.

[iv] Floricultural Crops

An area of 0.5 ha. has been demarcated for the demonstration of the commercial cultivation of floricultural crops. The crops include roses, jasmine, tube rose and bulbous plants.

Sylvisystems

Besides the conventional input systems for increased productivity, different systems of plantations have been evolved to increase the unit area productivity of land and obtain food, fuel, and fodder from the same piece of land. The Academy has established the following demonstrations for the benefit of the trainees and visitors coming from different parts of the country. The Sylvi systems that have been established are as follows :

- (i) **Agro-horticultural system** : This demonstration comprises of 150 m x 50 m plot planted with Balanagar variety of custard apple at the spacing of 5m x 2m and the area between the rows is being used for growing field crops like greengram, blackgram, cowpea etc.

This system will supply parallel production of fruit and field crops, which gives added income to the farmer.

- (ii) **Agro-forestry system** : It is a purely agricultural system in which tree component is introduced at appropriate spacing so that the available spaces can be utilised for growing food crops based on the principles of mutual benefit in terms of input utilisation and association.

This demonstration is in a block of 200m x 50m with alternate rows of *Leucaena leucocephala* (Lababool variety K-8) and *Acacia auriculiformis* planted at a spacing of 5m x 2m. Both these tree species are quick growing and drought tolerant. Cowpea variety T-86 is grown in the space between the rows.

- (iii) **Sylvi-Pastoral System** : This system has been designed to produce fodder for stall feeding/grazing purposes and supply of timber and fuel. An area of 180m x 50m unsuitable for cultivation has been planted with *Azadirachta indica* (Neem) plants at 5m x 5m, and the complete area is sown with *Stylosanthes hamata* which is a leguminous quick growing and drought tolerant fodder crop.

This system will demonstrate the effective utilisation of most unproductive land for economic benefit of farming community.

- (iv) **Horti-Pastoral System** : It is a flexible system in which unfertile and unproductive land can be utilised

for production of fruit crops and developed as a pasture either for grazing or stall feeding purposes. In this system *Embolica officinalis*, *Fesonia elephentian* and phalsa were planted with the space between the rows sown with *Stylosanthes hamata* and *Cenchrus ciliaris* in alternative patches. Nearly about four years all these fruits crops will start giving fruits and pasture also will be established for its full utilisation.

- (v) **Energy Planatation** : A block of 100m x 50m has been planted with a recommended variety of *Leucaena leucocephala* (Subabool) at 2m x 2m spacing. This high density plantation will be established to demonstrate the production of fodder and fuel in rural areas. This will encourage the small and marginal farmers to raise cattle and other animals, for milk, meat and drought purpose with the resources produced at their own farm.

Green belt and graded plantations: Apart from demonstrating all the above systems and latest technology in field and horticultural crops some areas particularly at boundaries of the campus were demarcated for growing graded plantation, which includes tree species of Eucalyptus, casurina, Tree Jasmine, and silver oak. In green belt areas different forest trees were closely planted which includes tree species like *Peltoforum*, *Siris*, *Gulmohar*, *Rain tree*, *Karanj*, *Casia sp.*, *Silk cotton*, *Mahwa*, *Ashoka*, *Neem*, *Soapnut*, *Drum Stick*, *Erythrina*, *Spathodia*, *Glyricidia*, *Parkia*, *Casurina*, *Thespesia*, *Tamarind*, *Jamun*, *Silver Oak*, *Acasia* etc..

NON-CONVENTIONAL ENERGY SYSTEMS

- [i] **Domestic Bio-Gas Plant** : A domestic bio-gas plant developed as a semi urban unit by Dr. K.G. Gollakota (Formerly of the A.P. Agricultural University) is under continuous demonstration at the Academy. In this unit non-edible oil cakes namely castor, karanj, neem or mahua etc. are used to produce the combustible gas i.e. Methane, which is fed into a store, which is ultimately connected to a burner for day to day use.
- [ii] **Wind Mill** : A Wind mill has been installed in the campus to harvest the wind energy for running a water pump. The wind mill has been installed with the guidance of NEDCAP. The harvested water is being utilised for surrounding avarues and plantations.

- [iii] **Solar water heater** : A multi-flat collector water heating system has been installed in the Academy for supplying hot water at 65°C. This hot water is being utilised for cooking and cleaning utensils in the kitchen room of Halls of Residence.

Consultancy and Advisory Services : The unit has been providing consultancy and advisory services in landscaping and ornamental horticulture to the nearby Government Institutes such as S.V.P. National Police Academy, Indian Grain Storage Research Institute.

[ii] PRODUCTION AND PUBLICATIONS

The Academy has taken up several publications on its own to help the research managers in keeping themselves in touch with the latest techniques of management. The Academy is publishing the materials developed by the Faculty for training purposes. Some of the publications are mentioned below:

1. Course materials and Reading materials for each course conducted by the Academy.
2. Case materials prepared on the basis of research studies undertaken by the faculty of NAARM.

GREEN AND GLORY

Green and Glory is a laboratory journal published for every Foundation Course based on the thesis of the trainees of the particular batch. For every issue 400 copies of the journal are printed and the copies are sent to all IAR Institutes. Agricultural Universities and user agencies like various development departments of agriculture, animal sciences, dairy corporation and fisheries departments of the various states. Dr.VE. Sabarathnam is the Chief-Editor of the journal.

NEWSLETTERS

NAARM News : 'NAARM News' brings out the news and information about the events, programmes, future programmes and the welfare activities of the staff to the Academy and is published every quarter. Dr.Devesh Kishore is the Editor in Chief of this Newsletter.

NAARM Alumni News : 'NAARM Alumni News' is a newsletter which publishes the information about the Alumni half yearly. Dr.R.K. Samanta is the Chief Editor of this Newsletter.

(iii) ADMINISTRATION AND ACCOUNTS

Introduction

For providing effective administrative services and in accounts matters, this unit has been arranged in six sections viz.

1. Admn. I Section - dealing with establishing matters
2. Admn. II Section - dealing with Cash and Bills
3. Admn. III Section - dealing with Stores and Purchase, Transportation and Budget matters
4. Admn. IV Section - dealing with Dairy and Despatch
5. Admn. V Section - dealing with all training matters and compilation and publication of rules and regulations relating to scientific, technical, administration etc.
6. Audit & Accounts Section - dealing with all financial matters

Objectives, Tasks and Functions

The Unit provides administrative and financial support for the smooth conducting of various training programmes of the Academy in addition to the day to day administration of the Academy and the NAARM Faculty.

Activities during the year 1987

The necessary administrative financial support was provided for the conducting of the Foundation Courses on Agricultural Research Management, Orientation Courses on Agricultural Research Management and a number of Workshops held during the year. The Unit also rendered necessary help in the administration of the Academy.

Dr.R.K.

3.2. STAFF POSITION

Sl. No.	Category	Sanctioned strength upto March, 1985			Additions proposed in VII Plan as recommended by IWSU
		Plan	Non plan	Total	
1.	Scientific	16	05	21	22
2.	Technical	12	03	15	38
3.	Administrative	17	12	29	24
4.	Auxiliary	05	04	09	13
5.	Supporting	10	07	17	41
	Total	60	31	91	138

3.3. FINANCIAL OUTLAY

(Rupees in lakhs)

Sl. No.	Head	VI Plan actuals		VII Plan requirement		Total
		Plan	Non-Plan	Plan	Non-Plan	
1.	Pay & Allowances	16.81	21.50	125.10	117.52	242.62
2.	T.A.	1.09	0.90	10.00	10.00	20.00
3.	Contingency	14.45	44.54	25.80	111.75	137.55
4.	Equipment	32.62	- -	62.50	- -	62.50
5.	Works	212.67	- -	277.10	- -	277.10
6.	Vehicles	4.55	- -	7.00	- -	7.00
7.	Library Books	11.94	- -	12.50	- -	12.50
8.	Others	- -	1.61	- -	10.73	10.73
Total		294.13	68.55	520.00	250.00	770.00

4. UNDP ADVANCED CENTRE ON AGRICULTURAL EDUCATION AND RESEARCH MANAGEMENT.

An advanced Centre on Agricultural Education and Research Management has been established at the Academy with assistance from UNDP/UNESCO/FAO with the primary objective of establishing a system for managing agricultural education and research so as to ensure continuous adjustments to changing national needs through scientific study of educational research management process and training key personnel in their application. The centre conducted one workshop on National Agricultural Research Projects to strengthen permanently the Capabilities of State Agricultural Universities to conduct location specific, production oriented research in the agro-climatic zones identified in their service areas and three courses on Agricultural Research Project Management; two courses on Human Resources Management and one course on Instructional Technology and Teaching Methods and another on Management Techniques for Information Handling in Technical Cells in ICAR Institutes, besides one International Seminar on 'Agricultural Research Systems and Management in the 21st Century'.

Dr. K.V. Raman, Director, NAARM, and Sub-Project Coordinator, Advanced Centre for Agricultural Education and Research Management, NAARM, visited U.K; U.S.A and Italy from May 4th to June 9th, 1987 to establish rapport with the institutions working in these areas and to identify persons who could assist in development of this programme as consultants in different areas. During his trip he visited a number of Universities, Institutes, Centres and interacted with many senior officials and educationists.

Two faculty members Dr. Devesh Kishore, Professor and Head, Educational Systems and Technology Unit and Dr. R.K. Samanta, Associate Professor and Head, Transfer of Technology Systems and Policies Unit, NAARM left of USA for advanced study for six months from September 24, 1987 in the areas of Educational Administration and Management, effective management of Agricultural education organisation, problems faced in implementation of curricula and development of new educational strategies and Audio-visual material production etc.

5. PHYSICAL FACILITIES

The Academy has set up the following laboratories to make the teaching-learning process more effective and help the trainees in effective use of mass media for dissemination of information.

- 5.1 **Closed Circuit Television Systems Laboratory:** It provides intensive training to scientists in preparation of effective and intelligible material for telecasting and to conduct role play exercises for personality development. The laboratory is also useful for developing proper speech communication skills and to improve the group communication process in building an efficient research team. Educational programmes related to structured learning are also being produced.
- 5.2 **Audio-visual Lab. including Photography unit :** This provides an excellent opportunity for scientists to have first-hand communication and practical experiences in preparation and use of A.V. aids for effective presentation.
- 5.3 **Speech and Mass Communication Laboratory:** It helps the Scientists to record their programmes such as interviews, discussion and talks specially from the rural listeners' point of view.
- 5.4. **Organisational Behaviour Laboratories :** These laboratories are organised for exercises in various aspects of organisational behaviour and help the trainees to understand human thought, action and behaviour.
- 5.5. **Self-Instructional carrels :** These carrels are being established at the Academy to provide an opportunity to the participants of various Training programmes, workshops, seminars and symposia in self-learning process. A slide projector, a video cassette player with monitor, micro-fich reader and an overhead projector will be made available in each carrel for this purpose.

The Academy has also established a Documentation and Information Systems on Agricultural Research Management. The Academy has, in addition, various types of Audio-visual equipments, such as Overhead Projector, Epidiascope, coloured transparency making unit, slide projector, 16mm projectors, tape recorders and public address system. The scientists are being trained in use of these equipment for effective presentation

of scientific material. All the lecture halls of the Academy, the Committee room and the conference hall are provided with the necessary audio-visual equipment to make the lecturing process more efficient & effective.

5.6. COMPUTER FACILITIES

The Academy provided with one Micro computer of UPTRON S-850 along with the Printer (Centronics 353) with 200 cps. It has the facilities such as the Data Analysis, Dataform, Graphics and Word Processing. During this year, the section acquired a PC-AT of DCM with 1 MB. System memory and one 40 MB Hard Disk drive along with a L & T Printer having a speed of 160 cps. These Computers are mainly used for Data Analysis of Research Projects and for Training. Further, they are also used for demonstration purpose in exposing the functions and operations of the PCs and Micro type computers to the ARS Trainees.

In addition, the computer cell has undertaken the following works during the year :

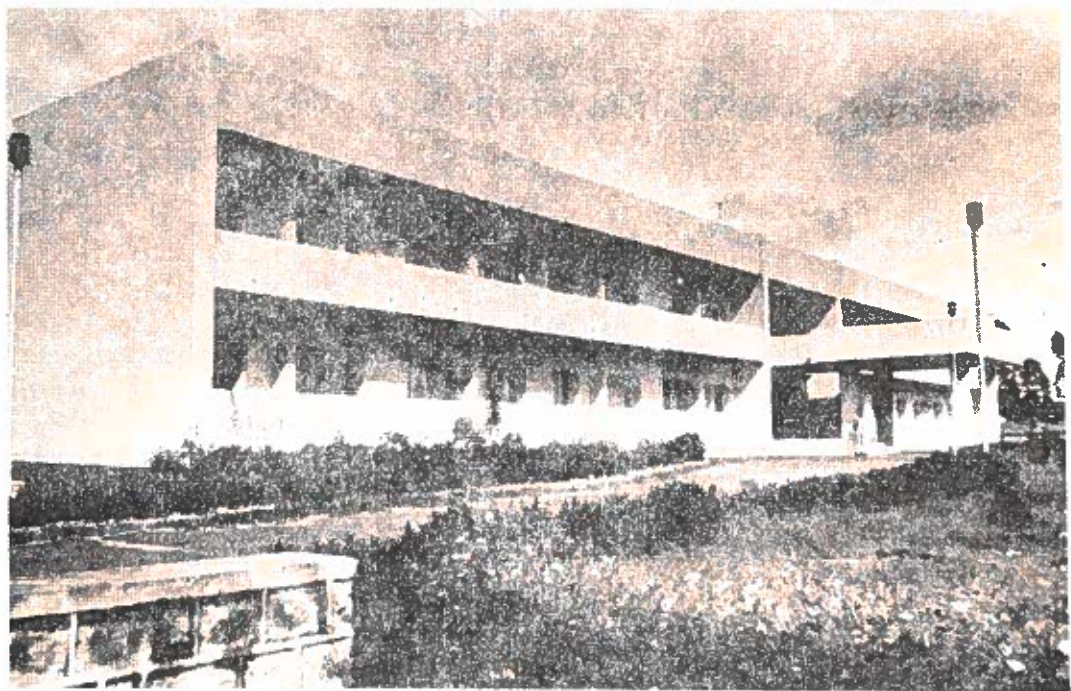
- i. Prepared the questionnaires and schedules for the data collection on various projects
- ii) Documented the Research Projects, Reports and Research Papers of various units, Bio-data of the Foundation Course Trainees attended during the year, and addresses of the members of the NAARM ALUMNI
- iii. Prepared the pay bills and pay slips of the Officers for every month
- iv) Analysed the data related to G.V.K. Rao Review Committee on ICAR functioning
- v) Bio-data of Faculty Members of NAARM have also been documented for ready reference.

5.7. BANKING FACILITIES

Central Bank of India, Golconda Fort Branch, Hyderabad has started Extension Counter (regular) at NAARM from November 10, 1987 to cater to the banking requirements of Academy Faculty Members, Staff and Trainees. The Extension Counter was inaugurated by Dr. K.V.Raman, Director, NAARM, in ground floor of the Administrative building.



Hon'ble Union Minister for Agriculture Sardar G.S.Dhillon inaugurating the Scientists' Home. Also seen Andhra Pradesh Chief Minister Shri N.T.Rama Rao and A.P.State Agriculture Minister Shri Rajagopal Reddy.



A view of the Scientist Home

HIGHLIGHTS OF THE YEAR 1987

6.1. DEDICATION OF THE ACADEMY TO THE NATION AND THE INAUGURATION OF BUILDING COMPLEX

The Union Minister for Agriculture and President Indian Council of Agricultural Research (ICAR), New Delhi, Sardar G.S. Dhillon inaugurated the Administrative building with air conditioned conference and committee halls; Academic block and Scientists Home and also laid down the foundation stone for the faculty building. While inaugurating the building complex Dr. Dhillon called upon scientists to function with high degree of social responsibility, service attitude and skill to face the challenges of the 21st Century.

Presiding over the function, Andhra Pradesh Chief Minister, Sri N.T.Rama Rao said that greater efforts were needed on the irrigation front. While a lot of water was available and irrigation projects planned, the water was not being utilised to the fullest extent. Research had to be done on the methods to utilise water to the drought prone areas. Shri Rama Rao inaugurated the Halls of Residence. Shri R.Rajagopal Reddy, Agricultural Minister, Andhra Pradesh and Dr. N.S. Randhawa, Director-General, ICAR, New Delhi also participated in the function. Speaking on the occasion, Dr.K.V.Raman, Director, NAARM, mentioned that Management holds the key for increased production for prosperity and the Academy will try to honour the commitment with zealousness and devotion.

Messages of Greetings and good wishes, wishing the function all success and the Academy continued progress and prosperity received from President Zail Singh, Prime Minister Rajiv Gandhi, A.P. Governor Kumudben Joshi, Minister of State Agriculture Yogendra Makwana and others.

An exhibition highlighting the various activities of NAARM was another attraction of the day. Through charts, posters, photographs and graphs, invitees had a glimpse of the organisation, objectives and achievements of the Academy.

6.2. NATIONAL WORKSHOP ON REVIEW OF AGRICULTURAL RESEARCH AND EDUCATION SYSTEMS IN INDIA

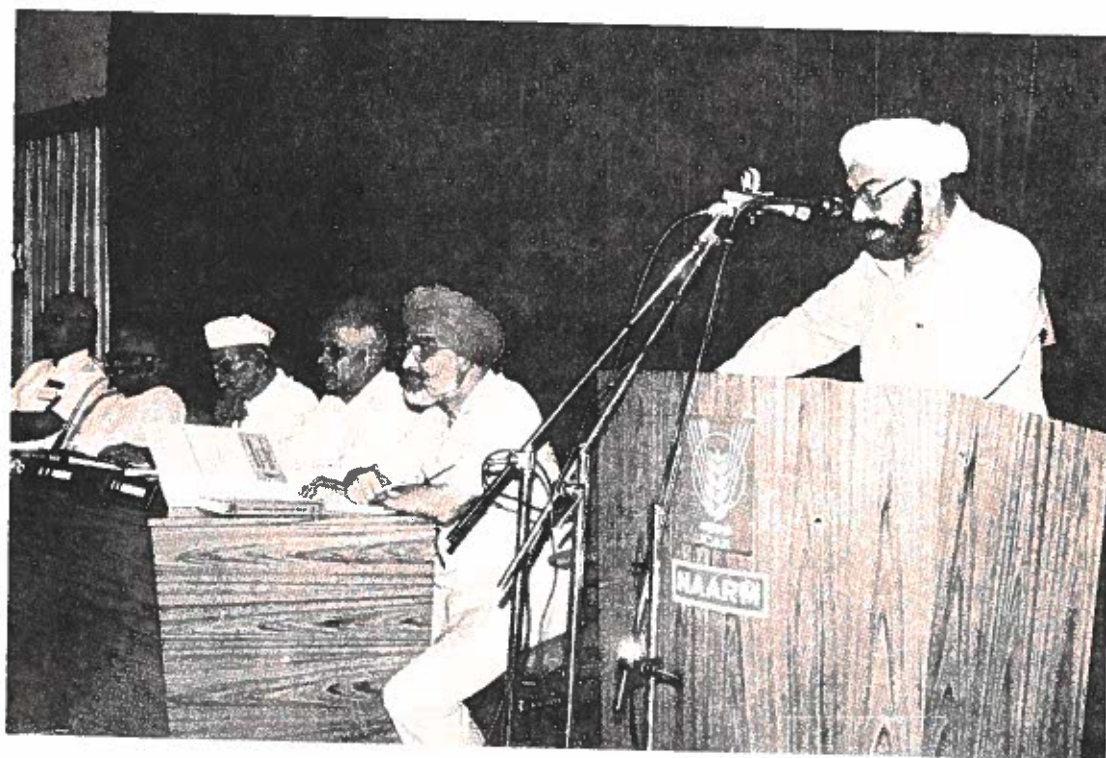
The ICAR Review Committee organised a National Workshop on 'Review of Agricultural Research and Education Systems in India' at NAARM, from October 9-11, 1987.

The Union Minister for Agriculture and President of ICAR, had constituted a committee under the Chairmanship of Dr.G.V.K.Rao, Former Member, Planning Commission, to look into the working of the ICAR and its Research Institutes. The Director of the Academy, Dr.K.V.Raman, functioned as its Joint Secretary. The main objective of the Committee was to review the existing organisational infrastructure, personnel policies and functional role of the ICAR system and its linkages with other related agencies in the light of the newer developments and challenges to Agricultural expected in the coming century and to suggest necessary means to equip the ICAR system adequately. In particular, the Committee had the following specific function to look into.

1. To review the existing role and organisational structure of ICAR and its institutional and the State Agricultural Universities and to suggest appropriate modifications with a view to meeting the new challenges of agricultural production and raising productivity economically.
2. To review the Institutional arrangements and to suggest improvements for augmenting national research facilities, revitalising education and reorienting extension education mechanism.
3. To review relative proportions of three kinds of research, viz. basic/fundamental, applied and operational/outreach.
4. To review the present linkages between agricultural research and extension and suggest arrangements for effective and expeditious application of new technologies for optimum utilisation of resources.
5. To review the role of ICAR in agricultural education through its own Institutions and the Agricultural Universities with special emphasis on raising standards of education and training.
6. To review the personnel policies with a view to attracting talent and building up requisite technical expertise especially in highly specialised but critical disciplines.
7. To review the linkages of ICAR with various client departments and other organisations to facilitate inter-institutional functioning in a complementary fashion with reference to national priorities.



Hon'ble Ministers being briefed about the layout of NAARM by
Dr.K.V.Raman, Director, NAARM.



Hon'ble Union Minister for Agriculture Sardar G.S.Dhillon inaugurating the National Workshop on Review of Agricultural Research and Education Systems in India. Also seen Dr. N.S.Randhawa, Director-General, ICAR; Shri A.P.Shinde, Former Union Minister of State for Agriculture; Prof. N.G.Ranga, M.P.; Shri C.Subramaniam, Former Union Agricultural Minister; Dr. G.V.K.Rao, Chairman, ICAR Review Committee.

8. To consider institutionalising the participation of the Department of Agriculture and Cooperation in developing the research programmes.

Apart from deliberating in detail on the various issues covered by its terms of reference, the committee had held discussions with (i) Directors of ICAR Institutes and some Senior Scientists, (ii) Vice-Chancellors of Agricultural Universities (iii) Secretaries of Central Government Departments of Agriculture, Cooperation and Rural Development/Councils etc. The Members of the Committee had also visited a number of ICAR Institutes, Agricultural Universities and related organisations and have held discussions in state capitals with Heads of Development Departments. The Committee had decided to hold a National Workshop on Agricultural Research and Educational Systems in India. The Committee had also circulated questionnaires to Directors of ICAR Institutes, Scientists in the ICAR organisations, Agricultural Production Commissioners in the State, as well as Vice-Chancellors of Agricultural Universities. The following discussion papers had been prepared based upon the deliberations and responses and had been compiled by the faculty of NAARM. In addition to these, some experts had also been requested to present invitational papers on specific topics. The discussion papers had been grouped under the following heads :

- A. ORGANISATIONAL STRUCTURE OF ICAR TO FULFIL ITS MANDATE
- B. ICAR INSTITUTES - THEIR ROLE AND RELATIONS WITH HEAD-QUARTERS
- C. PERSONNEL POLICIES OF ICAR
- D. ORGANISATION OF APPROPRIATE TECHNICAL AND ADMINISTRATIVE SUPPORT SERVICES
- E. LINKAGES OF ICAR WITH STATE AGRICULTURAL UNIVERSITIES
- F. ICAR AND EXTENSION ACTIVITIES
- G. RESEARCH PLANNING, IMPLEMENTATION, MONITORING AND EVALUATION IN THE ICAR

6.3. INTERNATIONAL SEMINAR ON AGRICULTURAL RESEARCH SYSTEMS AND MANAGEMENT IN THE 21ST CENTURY

The pace of agriculture is undergoing a rapid change and is moving from the pattern of subsistence farming to an organised industry. These changes, which are more evident in the developing countries, will become more significant as we enter the 21st Century. The demands on the National Agricultural Education, Research and Extension Systems will change rapidly and an appropriate systems of management and structural changes are imminent that will require vast adjustments in approaches and management. The Academy in collaboration with the Alumni Association of NAARM, organised an International Seminar on 'Agricultural Research Systems and Management in the 21st Century' from December 8-10, 1987 with the following objectives.

1. Reviewing the current pattern of Agricultural Research and Management Systems in the country.
2. Projecting the future demands of food, fibre and fuel for the 21st Century.
3. Developing technological innovations to meet the challenges of the 21st Century.
4. Developing appropriate National Agricultural Research Systems to meet the needs of the 21st Century.

The Seminar was organised under four specific themes. Under each theme, invited papers by eminent scientists were presented followed by other contributed papers. The papers were arranged according to the themes as outlined below :

Theme I dealt with projecting the future demands of food and fibre for the 21st Century. The increased demands arise not only from the increased population, but also from changes in economic status altering the consumption requirements. Four papers were presented in this theme, two concerning the food from plant sources, one on food from the animal sources and one on the demands of cotton fibre.

Theme II explored the technological innovations to meet the challenges of the 21st Century. Four papers were included under this theme. After discussing the current state of the technology, the first two papers, one on plant sciences and another on animal sciences. The third paper dealt with the prospects of biotechnology examining the provocative ques-

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Dr.K.V.Raman, Director, NAARM, welcoming the participants. Also seen [L to R] Dr.M.M.Anwer, General Secretary, NAARM Alumni Association; Dr.N.G.P.Rao, Chairman, A.S.R.B. and Dharni P.Sinha, Principal, Administrative Staff College of India, Hyderabad.

tion "Is Bio-technology an answer to our future needs ?". The fourth paper examined the future priorities of agricultural research.

Theme III reviewed the current agricultural research and educational systems in the country. The first paper presented an overview of the Agricultural Research System in India. The second paper reviewed the agricultural universities system in India. The third paper discussed the role of the All India Coordinated Research Projects which have been unique and successful Projects in developing technologies for increased production.

Theme IV dealt with the development of appropriate Agricultural Research and Educational Systems to meet the needs of the 21st Century. Six papers outlined this issue. These papers made suggestions on the development of National Research and Education Systems, that are responsive to the priority setting in research, and the educational and training needs of the 21st Century.

Dr.Dharni P. Sinha, Principal, Administrative Staff College of India, Hyderabad, inaugurated the Seminar. Dr.N.G.P. Rao, Chairman, Agricultural Scientists Recruitment Board presided over the Seminar, on December 8, 1987.

The first session was chaired by Dr.A. Appa Rao, Vice Chancellor, Andhra Pradesh Agricultural University, Hyderabad. A paper entitled 'Perspectives in Agricultural Development' was presented by Dr.S.S.Khanna & M.V. Pavate Adviser (Agriculture) and Joint Adviser (APS) respectively of the Planning Commission, New Delhi.

The Second session was chaired by Dr.R.P.S.Tyagi Member, Agricultural Scientists' Recruitment Board, New Delhi. Two international papers entitled 'Implications of Bio-Technology in Agriculture' and 'Expected Technological advances to meet the needs of the 21st Century - Focus on Animal Sciences' by Dr.V.L. Chopra, IARI, New Delhi and Dr.R.M.Acharya, Dy. Director-General, ICAR, New Delhi respectively. Another contributed paper namely 'Educational and Training needs to meet the Agricultural Production requirements of the 21st Century' was presented by Dr.Dan Aldrich, University of California in the afternoon session.

The Third session was chaired by Dr.J.S.P. Yadav, USAID/ American Embassy, New Delhi. One invitational paper entitled 'Is Bio-Technology an answer to our future needs ?' was presen-

ted by Dr. Karl Maramorosch, Rutgers University, N.J., USA and another paper entitled 'Private Sector role in agricultural research - Prospects and Projections ' was presented by Dr. Donald Duvick Pioneer Hi-Bred, International Inc., USA.

The Fourth session was chaired by Mr. Charles Antholt, Assistant Director of Agriculture, USAID/American Embassy, New Delhi. Three invitational papers entitled 'Expected technological advances to meet the needs of the 21st Century - Focus on Agriculture' by Dr. N.S.Randhawa, Director-General, ICAR, New Delhi and 'Priority setting for agricultural research' by Dr.Edward D. Knipling, Dy. Administrator, U.S. Dept. of Agriculture, Washington were presented.

The Fifth session was chaired by Dr.N.S. Randhawa, Director-General, ICAR, New Delhi. One paper entitled 'Global food production strategy' was presented by Dr.Bentley, O.G., Asst. Secretary for Sci. and Edn. U.S. Dept. of Agriculture (USDA), Washington.

The Sixth session was chaired by Dr.K.Mohan Naidu, Director, Sugarcane Breeding Institute, Coimbatore. Two papers entitled 'Technological innovations in Crop Sciences to meet the challenge of 21st Century : Multiple cropping approach' by Dr.S.P. Ghosh and 'Projections for the quantitative and qualitative requirements of cotton fibre for the 21st century and the genetic approaches to achieve them' by Dr.S.S. Narayanan, Head (Crop Improvement) Central Inst. for Cotton Research, Nagpur.

The Seventh Session was chaired by Dr. A.B. Joshi, Pune. Three invitational papers were presented namely 'Agricultural Research Systems in India' by Dr. T. Balaguru, 'Management of All India Coordinated Research Projects and their role in National Agricultural Research System' by Dr.K.Venka-teswarlu, Dr.V.Raghavendra Rao and Dr.K.V.Subba Rao of NAARM and 'Agricultural Universities Systems in India' by Dr.Sukhdev Singh, Vice-Chancellor, PAU, Ludhiana.

The eighth session was chaired by Dr.Guy Baird. One invitational paper entitled 'National Agricultural Research Systems in the Future' was presented by Dr. K.V.Raman, Director, NAARM and one contributed paper entitled 'Need for an appropriate National Agricultural Research System for meeting the challenges of the 21st Century' by Dr.S.S. Narayanan, Central Institute for Cotton Research, Nagpur.

Mr.R.P. Kapoor, Director-General, National Institute of Rural Development, (NIRD), Hyderabad chaired the concluding

session. The International Seminar on Agricultural Research Systems and Management in the 21st Century was successful in focussing attention of the scientists to the challenging tasks of increasing the production of food, fuel and fibre through optimising production using existing technologies, identifying gaps and lapses and developing research strategies. The various themes that were discussed in the Seminar were the projection of the future demands of food and fibre, development of technological innovation, review of the current agricultural research and educational systems in the country and the development of appropriate national agricultural research systems to meet the needs of the 21st Century. It was felt that such seminars would contribute towards improving agricultural production in the country.

One pre-seminar publication incorporating 16 invited and contributed papers to the seminar was brought out. The Academy and the NAARM Alumni Association proposed to bring out a post-seminar volumen also which will incorporate all the papers presented and discussed in the Seminar.

One hundred and fourteen distinguished Agricultural Scientists from abroad and eminent Indian Agricultural Scientists, Policy Makers, Administrators, Senior Functionaries from ICAR and the Agricultural Universities participated in the Seminar.

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7. NAARM ALUMNI ASSOCIATION

The General Body meeting of the NAARM Alumni Association was held on 8-12-1987 at National Academy of Agricultural Research Management, Hyderabad, under the Presidentship of Dr.K.V. Raman, and was attended by 103 members of the Association. The President requested the Regional Representatives to take more interest in organising regional activities on behalf of the Alumni Association and to organise regional get togethers, Seminars and Symposia. The total membership of the Association was increased to 448 members as on 8-12-1987. The President made the following suggestions to the members :

1. NAARM Alumni may join together for undertaking Rural Development Programmes all over the country around their place of work.
2. The Alumni Association may conduct bi-annual seminar coupled with a General body meeting.
3. The Alumni may interact to a greater extent through the newsletter.
4. The Field Experience Training information gathered by the trainees of NAARM may be consolidated and published in an appropriate form.
5. The Executive Committee may bring out an upto date Directory of the members to enable them to correspond with one another

Consequent upon the retiring of Mr.P.V.R. Murthy, the Ex-Officio Treasurer of the Association, Mr.M.P. Chandrasekharan, Accounts Officer, NAARM, has assumed the office.

PUBLICATIONS

8 1. Scientific Papers Published

Rajagopalan M. and Balaguru, T. (1987) Managerial Styles of ICAR Directors through grid analysis. Decision 14(1) : 9-12.

Sabarathnam VE. (1987) Extension Education Institute "My Alma Mater" in the Souvenir of Silver Jubilee Celebration of Extension Education Institute of Andhra Pradesh Agricultural University, Hyderabad organised in October 1987, pp12-13.

Samanta, R.K. (1986) Effective Agricultural Communication Strategy : Need for Efficient Transfer of Agricultural Technology, *Bangladesh Journal of Extension*. 1(2) : 91-100.

Samanta, R.K. (1987) Extension Training and Education need for Effective Transfer of Technology in North Eastern Hills Region, *Journal of North-East India Council of Social Science Research*, Vol. 11 (I) 29-32.

Samanta, R.K. and M.M. Adhikary (1987) Scientists' Perception of Communicative Openness in Agricultural Research Organisations, *Decision*, Vol. 14 (I) : 19-24.

Samanta, R.K. (1987) Diverse Marriage Systems among Tribes of North-East India, *Man in India*. 67(3).

8.2. Papers presented at National and International Symposium/Seminar/Conference/Workshops

Balaguru T. and Rajagopalan, M. 1987. Training Needs of Agricultural Research Scientists on Computer use. Presented at the Symposium on Research Data Processing and Information Systems in Agricultural held at IASRI, New Delhi, May 25-26, 1987.

Balaguru, T. and Raman K.V. Agricultural Research Systems in India on 'Agricultural Research Systems and Management in the 21st Century' presented at International Seminar on "Agricultural Research Systems and Management in the 21st Century" organised by NAARM Alumni Association, Rajendranagar, Hyderabad from December 8-10, 1987.

Raman, K.V. Scientists' Training Experiences in Promoting Interactions with the Farmers. Presented at the International Workshop on 'Farmers and Agricultural Research : Complementary Methods' held at the University of Sussex Brighton, England from July 27-31, 1987.

Raman, K.V. 'Role of National Agricultural Research Systems in Policy Planning for Agricultural Research'. Presented at the National Symposium on 'Impact of Agricultural Research and its Management in Accelerating Socio-economic Transformation' organised by Agricultural Research Service Scientist's Forum and Indian Council of Agricultural Research, New Delhi, from 19-21, September, 1987.

Raman, K.V. National Agricultural Research Systems in the future at International Seminar on 'Agricultural Research Systems and Management in the 21st Century' organised by NAARM

Alumni Association, Rajendranagar, Hyderabad from December 8-10, 1987.

Raman, K.V. and Balaguru, T. 1987. NARP - An Innovative Approach towards FSR in India. Presented at the Workshop on Management of Crops, Livestock and Environment - A Farming Systems Perspective held at NIRD, Hyderabad, January 19-22, 1987.

Venkateswarlu, K. presented the following three papers in the National Seminar on Training Strategies for Human Resource Development in Agriculture from October 29-30, 1987, organised by Extension Education Institute, (APAEI), Hyderabad in collaboration with Directorate of Extension Min. of Agriculture (Dept. of Agriculture & Cooperation), New Delhi and Andhra Pradesh Chapter of Indian Society of Extension Education, Hyderabad.

- i. Extension Management Using PERT Network.
- ii. Training Needs of Agricultural Scientists of ICAR and Agricultural Universities.
- iii. Hierarchical Model for Designing a Training Programme.

Venkateswarlu, K. "Project Management Techniques" - Paper presented in Symposium 'The Role of Scientific Research and its Management in Accelerating Socio-economic Transformation' held during September 19-20, 1987 at IARI, New Delhi.

Venkateswarlu, K. Subba Rao, K.V.; Raghavendra Rao, V. Management of All India Coordinated Research Projects and their role in National Agricultural Research Systems' at International Seminar on 'Agricultural Research Systems and Management in the 21st Century organised by NAARM Alumni Association, Hyderabad from December 8-10, 1987.

8.3. SPECIAL LECTURES DELIVERED

Dr. Devesh Kishore

- ** Communication and Media Management in the National Workshop on Management of Agricultural Extension for Oilseeds production at National Institute of Rural Development, Hyderabad.
- ** Issues and Problems in training adults at National Institute of Small Industry Extension Training, Yousufguda, Hyderabad.



NAARM Faculty

Sitting (L to R)

: 1. Dr. K. Venkateswarlu, 2. Dr. V. Raghavendra Rao 3. Dr. Devesh Kishore 4. Dr. K. V. Raman 5. Dr. K. V. Subba Rao 6. Dr. M. P. Singh 7. Dr. S. N. Saha.

Standing Row (L to R)

: 1. Sh. R. B. Gaddagimath 2. Sh. M. P. Chandrasekharan 3. Dr. V. E. Sabarathnam 4. Dr. K. P. C. Rao 5. Dr. M. M. Anwer 6. Dr. P. Manikandan 7. Sh. Chittirai Chelvan 8. Dr. K. Pratap Reddy 9. Dr. R. K. Samanta 10. Dr. T. Balaguru 11. Sh. Mohd. Habeebuddin.

- ** 'Instructional System Design and Method' and 'Model approach in design and modification of curriculum at microlevel' at National Institute of Rural Development, Hyderabad.
- ** Communication - Individual oriented personal contact at State Bank Institute of Rural Development, Hyderabad.
- ** 'Formal and Informal Communication', Aids and Barriers to effective Communication', at S.V.P. National Police Academy, Hyderabad.

Dr.K. Venkateswarlu

- ** 'R & D Planning for large scale food industry to Tomato Products' for Senior Managers of Agro-Industries at National Institute of Rural Development, Hyderabad.

Shri K. Pratap Reddy

- ** 'Leadership Styles' at the Training Programme for Asst. Directors & Deputy Directors of Agriculture incharge of T & V System at Extension Education Institute, Andhra Pradesh Agricultural University, Rajendranagar, Hyderabad.

Dr.M.M. Anwer

- ** 'Human Behaviour' in the training programmes of S.V.P. National Police Academy, Hyderabad.

9. MISCELLANEOUS

9.1. Faculty Development

- ** Dr.K. Venkateswarlu, Professor; Dr.P. Manikandan, Associate Professor and Mr.R.B.Gaddagimath, Librarian attended a training programme on Computer awareness for government informatics from April 20-24, 1987 at National Informatics Centre, Hyderabad.
- ** Shir K. Pratap Reddy, Associate Professor attended a short course on 'Introduction to Personal Computers' from August 26-29, 1987 conducted by National Informatics Centre (Southern Region) at Hyderabad.
- ** Mr.R.B.Gaddagimath, Librarian, NAARM, attended Workshop on Information Management Techniques at Indian Institute of Technology, Bombay from June 1-12, 1987. This programme was sponsored by Department of Scientific and Industrial Research, Government of India.

9.2. FOREIGN VISITS

- ** Dr.K.V.Raman, Director, NAARM, and Sub-Project Coordinator, Advanced Centre for Agricultural Education and Research Management, NAARM, visited U.K., U.S.A. and Italy from May 4th to June 9th, 1987 to establish rapport with the Institutions working in these areas and to identify persons who could assist in development of this programme as consultants in different areas. During this trip he visited number of Universities, Institutes, Centres and interacted with many senior officials and educationists.
- ** Dr.K.V.Raman, Director, NAARM, was sponsored by the Indian Council of Agricultural Research, New Delhi to attend the workshop on 'Farmers and Agricultural Research : Complementary Methods' held at the Institute of Development studies at the University of Sussex Brighton, England from July 27-31, 1987. He presented the paper entitled "Scientists' Training Experiences in Promoting Interactions with the Farmer". Dr.Raman interacted with the Heads of the Departments at the University of Sussex Brighton, England.
- ** Dr.Devesh Kishore, Professor and Head, Educational Systems and Technology Unit, NAARM, left for USA for advanced study for six months from September 24, 1987

In Educational Administration and Management at University of Arizona, under the Fellowship from the Advanced Centre for Agriculture Research and Educational Management Project funded by FAO/UNDP.

- ** Dr.R.K. Samanta, Associate Professor and Head, Transfer of Technology Systems and Policies Unit, NAARM left for advanced study on Audio Visual material production for six months at Cornell University, Ithaca, USA on September, 24 under a fellowship from the Advanced Centre for Agriculture Research and Educational Management Project funded by FAO/UNDP.

2.3. DISTINGUISHED VISITORS

- ** A five-member delegation from the USSR, consisting of Agricultural Scientists and Researchers under the leadership of Mr. Balandin Y.N., Dy. Chairman of USSR AGROPROM visited NAARM on February 10, 1987 to explore areas of common interest in agriculture and farm production for an agreement valid till 2000 AD. The team consists of :

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| Academician
Kashtanov A.N. | - Vice-President of All Union Scientific Agricultural Academy named after V.I. Lenin. |
| Academician
Yarnykh V.S. | - Director of All-Union Research Institute of Veterinary Sanitation |
| Mr.Gureev N.N. | - Director of All-Union Association 'Agricultural and Industrial Export' |
| Mr.Popov N.N. | - Senior Specialist of the Department of Foreign Relations |
| Miss Kalashni
Kova O.P. | - Interpreter |

The main efforts of the delegation were focussed at identifying possible areas of interest and formulating appropriate project during the visit.

- ** Tony Williamson, Don Snowden Centre Memorial University of New Brunswick, Canada visited the Academy on January 15, 1987.

- ** A.W.Qureshi, Animal & Poultry, FAO, Rome and Dr.A.W.Vanden Ban, Foulkesweg 82 A 6703 BX Wageningen, Netherlands visited the Academy on February 17, 1987.
- ** Godson D. Ononum, Fed. Agric. Coordinating Unit, Nigeria, visited the Academy on March 20, 1987.
- ** Dr.Aregay Waktola, Research and Publication Officer, Addis Ababa University and Satish Mishra, Ethiopia visited the Academy on April 4, 1987.
- ** Mr. Daud. Mohemad and Eltoyab Ibrahim Aldelkoin, Agricultural Bank of Sudan visited the Academy on April 8, 1987.
- ** Sri S. Sundararajan, Commissioner, Agricultural and Rural Development, Govt. of Orissa, Bhubaneswar, visited the Academy on May 14, 1987.
- ** Mr.Norman Upoff, Rural Development Commissioner, Cornell University, Ithaca, U.S.A. visited the Academy on June 30, 1987.
- ** William B. Lacy, USAID and Paul B. Thompson, USAID visited the Academy on 5th and 8th August, 1987 respectively.
- ** J. Lin Compton, Cornell visited the Academy on August 12, 1987.
- ** Dr.S.W. Eremi, Dr.RPA Unamma, Mr. I.B. Edekobi and Mr. I. Aerbaore, Federal Agric. Coordinating Unit, Ibadan, Nigeria visited the Academy on September 14, 1987.
- ** The following nine member team of Agricultural Scientists from African Countries visited the Academy on September 15, 1987.
 1. Dr. Adomou Moustapha, Director, INAR Research Station B.P. 03, N'Dzlee N'dali, Benin.
 2. Dr. Andre Jules Bayogo, Director, Agric. An. Husbandry & Vet. Res. Min. of Rural Development, B.P. 786, Bangui.
 3. Dr.Bichara Cherif Daoussa, Director, Min. of Agric. & Rural Dev., B.P.411, N'D Jamena, CHAD.

4. Dr. Koffi Goli, Research Director, Instt. of the Savannas (IDESSA) B.P.663, Bouake, COTE D'IVOIRE.
 5. Dr. Essima Osse Michel, Director, Instt. of Agronomy and Forest Res. (IRAF), National Scientific and Technology Research Centre (CENARESI) B.P. 2246, Libereville, GABON.
 6. Dr.Diamoutene Dotienga, Dy. Director-General, Institute of Rural Economics (IER), Div. of Agric. Res. Ministry of Agriculture, B.P. 258, Bamako, MALI.
 7. Dr. Idrissa Soumana, Director-General, National Agric. Res. Instt. of Niger (INRAN), B.P. 429, Niamey, NIGER.
 8. Prof. J.Y. Yayock, Director, Instt. for Agric. Research Ahmadu Bello University, P.M.B. 1044, Zaria, NIGERIA.
 9. Dr. Leopold Gahmanyi, Director, Instt. of Agric. Sci. of Rwanda (ISAR), Station of Rubon, B.P. 138, Butare, RWANDA.
- ** Shri T. Hanumantha Rao, Director-General (Engineer-in-Chief), Water and Land Management Training and Research Institute (WALAMTARI), Himayatsagar, Hyderabad visited the Academy on October 16, 1987..
- ** Dr.K.W. Riley, Sr. Programme Officer, IDRC, New Delhi visited the Academy on October 30, 1987.
- ** A Team of Credit Supervisors of the Pilot Rural Credit Project of National Savings Bank, Colombo (Sri Lanka) visited the Academy on Nov. 11, 1987.
- ** Shri D.R.Hegde and Shri D.S.Patil, Agricultural Officers (SMS) Information and Training, Sirsi, Karnataka State and Shri Balaram Bal, Sr. Research Assistant, Soil Testing Laboratory, Bhawanipatnam, Orissa, visited the Academy on Nov. 20, 1987.
- ** Dr.V.N. Shukla, Vice-Chancellor, Panjabrao Krishi Vidya-peeth, Akola, visited the Academy on December 19, 1987.
- ** Mr. Styrbjorn Alstrom, Scientist, Dept. of Plant Husbandry, Swedish University of Agricultural Sciences, Uppsala, Sweden, visited the Academy on December 23, 1987.

9.4. HONOURS AND AWARDS

NAARM BAGGED THE "FACT FERTILIZER COMPANY'S SHIELD" IN ROSE SHOW

Academy's Rose Garden was adjudged the best Institutional Garden in the twin cities (Hyderabad & Secunderabad). The Academy bagged the prestigious 'FACT' Fertilizer Company's shield for this. NAARM also displayed the best Indian breed floribunda roses and won both the first prizes for these classes. In addition to these the Academy won seven second prizes and five third prizes. The Academy thus got a total of fifteen prizes winning the second highest aggregate points among all the competitors in the show.

ROSE BLOOM AT NAARM

The Academy bagged the Horticultural Society of Secunderabad Challenge Cup for the Queen of the show, at the First Flower show held at YMCA, Secunderabad on December 6, 1987. The Academy's Rose Garden won the best institutional garden rolling cup. Also NAARM bagged the four first prizes in the different rose colours and eight second prizes in different varieties.

9.5. IMPORTANT COMMITTEES OF THE ACADEMY

9.5.1. Management Committee

The Academy is guided in planning, organising and conducting of various types of training programmes and research studies by the Management Committee of NAARM. The Management Committee approves the programmes to be undertaken. The Academy ensures their implementation both in terms of physical targets and schedules and provides broad guidelines for future programmes. The Management Committee also helps in formulation and consideration of proposals for Five Year Plan and Annual Plan and takes periodical review of progress of development schemes. The composition of the Management Committee of NAARM is as follows :

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|----|---|----------|
| 1. | Dr.K.V.Raman
Director, NAARM | Chairman |
| 2. | Dr. Baidyanath Misra
Deputy Chairman
State Planning Board
Orissa Secretariat
Bhubaneswar. | Member |



Work on Computer in progress



Prizes bagged by NAARM

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|-----|--|--------|
| 3. | Dr. B. Kalia
Vice-Chancellor,
Bikaner, Jaipur. | Member |
| 4. | Dr. B. Panda
Director,
Central Avian Res. Institute. | Member |
| 5. | Dr. T.P. Ojha,
Director,
Central Institute of
Agricultural Engineering,
Bhopal. | Member |
| 6. | Dr. A.Appa Rao
Vice-Chancellor,
A.P. Agricultural University,
Hyderabad. | Member |
| 7. | Dr. I.P. Abrol,
Dy. Director General (SAE)
Indian Council of Agril. Res.
New Delhi. | Member |
| 8. | Dr. R.S. Paroda
Director,
National Bureau of
Plant Genetic Resources
New Delhi. | Member |
| 9. | Dr. K. Hanumantha Rao
Director,
Animal Husbandry,
Government of A.P.,
Hyderabad. | Member |
| 10. | Director of Agriculture
Government of Punjab,
Chandigarh. | Member |
| 11. | Dr. Maharaj Singh,
Dy. Director General (Educ.),
Indian Council of Agric. Res.
New Delhi. | Member |
| 12. | Sri. S.W. Oak,
Director (Finance),
Indian Council of Agril. Res.
New Delhi. | Member |

13. Shri G.C.Sharma,
Sr.Admn.Officer,
NAARM, Hyderabad.

Member-Secretary

The Management Committee besides assisting Director of the Academy in the discharge of his functions particularly concentrates his attention on the Research and other programmes of the Academy and ensures their implementation both in terms of physical targets and time schedules, while examining these programmes periodically, pinpoints bottlenecks, if any, and suggests suitable remedial measures.

The Eleventh Meeting of the Management Committee of the Academy was held on April 16, 1987.

9.5.2. STEERING COMMITTEE ON ADVANCED CENTRE ON AGRICULTURAL EDUCATION AND RESEARCH MANAGEMENT

This Committee was initially called the Sub-Project Advisory Committee and has been later renamed as Steering Committee. The Committee will periodically review the progress and plan of the project and advise the project staff. This Committee will meet at least once in a year and while reviewing the progress of the projects will provide broad guidelines for its implementation based on the suggestions of user agencies. The following are the members of the Steering Committee.

- | | | |
|----|---|----------|
| 1. | Dr.K.V.Raman,
Director, NAARM. | Chairman |
| 2. | Dr.Maharaj Singh,
Dy. Director General (Edn),
ICAR, New Delhi. | Member |
| 3. | C.. K.Anant Rao,
Former DDG (Edn), ICAR,
and Former Director,
NAARM. | Member |
| 4. | Dr.Guy Baird,
Indian Coordinator,
WINROCK International
Support Services Contract
Chanakyapuri,
New Delhi. | Member |
| 5. | Dr.A.Appa Rao,
Vice-Chancellor,
APAU, Hyderabad. | Member |

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| 6. | Dr. A.M. Michael,
Director,
Indian Agril. Res. Instt.
New Delhi. | Member |
| 7. | Dr. R. Nagardenkar,
Director,
National Dairy Res. Instt.
Karnal. | Member |
| 8. | Dr. A. Ram Mohan Rao
Dean, PG Studies,
A.P. Agril. University,
Hyderabad. | Member |
| 9. | Dr. Devesh Kishore,
Professor, NAARM,
Hyderabad. | Member |
| 10. | Dr. V. Raghavendra Rao,
Professor, NAARM,
Hyderabad. | Member |
| 11. | Sh. G.C. Sharma,
Sr. Admn. Officer,
NAARM, Hyderabad. | Member-Secretary |

The first meeting of the Steering Committee on the Advanced Centre on Agricultural Education and Research Management was held on 31st January, 1987 at NAARM, Hyderabad.

10. PERSONNEL

Dr.K.V.Raman Director

Director Personal Cell

- | | | |
|----|-------------------|------------------|
| 1. | Sh. P.P. Krishnan | Sr. Stenographer |
| 2. | Sh. T.Sreenivas | Jr. Clerk |
| 3. | Sh. P. Swamy | SSG I |

I. Agricultural Research Systems and Policies Unit

- | | | |
|----|--------------------|---|
| 1. | Dr.M. Rajagopalan | Professor, Unit-in-Charge, (Retd. on Sept 30, 1987) |
| 2. | Sh.VE. Sabarathnam | Assoc. Professor |
| 3. | Dr.T.Balaguru | Assoc. Professor |
| 4. | Dr.A.Bandyopadhyay | Assoc. Professor (Joined on 1.9.1987) |
| 5. | Sh.M.J. Pradeep | Computer Operator (Joined on 18.11.1987) |
| 6. | Sh. J.S.R. Murthy | Stenographer |

II. Educational Systems and Technology Unit

- | | | |
|----|-----------------------|---------------------------|
| 1. | Dr.Devesh Kishore | Professor, Unit-in-Charge |
| 2. | Dr.M.P. Singh | Professor |
| 3. | Sh. N.R.Nageswara Rao | A.V.Operator |
| 4. | Sh.K.S.Sharma | Stenographer |

III. Transfer of Technology Systems & Policies Unit

- | | | |
|----|-----------------|--|
| 1. | Dr.R.K. Samanta | Assoc. Professor Unit-in-Charge |
| 2. | Sh.R.C.Chelvan | Assoc. Professor (Joined on 28.5.1987) |

VII. Centralised Services

(i) Farm Operation Unit

- | | | |
|----|---------------------------|---------------------------------------|
| 1. | Mr.V.Murali | Farm Manager |
| 2. | Sh.Mohd. Abdul Basith | Lab. Technician |
| 3. | Sh.K.Rangaswamy | Field Assistant |
| 4. | Sh.B.Veeriah | Farm Asstt. |
| 5. | Sh.Ahire Laxman
Maharu | Farm Asstt. (Joined
on 30-11-1987) |
| 6. | Sh.K.Prabhudas | Jr. Clerk |
| 7. | Sh.K.Thirumalaiah | Pump House Operator |
| 8. | Sh.K.Obulapathi | Pump House Operator |

(ii) Production and Publication

- | | | |
|----|------------------------|---|
| 1. | Sh.R.V.V.S.Prakasa Rao | Asst.Editor |
| 2. | Sh.A.Murali Mohan Rao | Artist |
| 3. | Sh.K.Desai Babu | Dark Room Asstt. (Joined
on 5-12-1987) |
| 4. | Sh.S.Swamy | SSG-III |
| 5. | Sh.C.Chandramouli | SSG-I |

(iii) Administration and Accounts

- | | | |
|----|-------------------------|--|
| 1. | Sh.G.C.Sharma | Sr.Admn.Officer, Unit-in-
Charge |
| 2. | Sh.M.P.Chandra Sekharan | Accounts Officer (Joined
on 7-8-1987) |
| 3. | Sh.Mohd.Habeebuddin | Asst.Admn.Officer. |
| 4. | Sh.D.Venkateswarlu | Jr. Hindi Traslator |

5.	Sh. K. Lakshmi Ramaiah	Superintendent
6.	Sh. J. D. Sastri	Superintendent (on deputation joined on 10-12-1987)
7.	Sh. T. A. Subhik Kumar	Assistant
8.	Sh. Y. Sankara Rao	Assistant
9.	Sh. R. Shankar Narayan	Assistant (Joined on 6-2-1987) (Relieved on 31-12-1987)
10.	Sh. A. P. Ramachandran	Assistant (on deputation joined on 26-11-1987)
11.	Sh. P. P. Brahmaji	Sr. Clerk
12.	Sh. C. Bagaiah	Sr. Clerk
13.	Sh. P. G. Kohad	Sr. Clerk
14.	Sh. W. S. Bhat	Sr. Clerk
15.	Sh. V. Murali Krishna	Jr. Stenographer (relieved on 4-2-1987)
16.	Sh. M. Venkatesh	Jr. Stenographer
17.	Sh. N. Raghunath	Jr. Stenographer (Joined on 22-8-1987)
18.	Sh. K. Prabhudas	Jr. Clerk
19.	Sh. M. Narasimha Rao	Jr. Clerk
20.	Sh. T. Srinivas	Jr. Clerk
21.	Sh. M. Dinesh	Jr. Clerk
22.	Sh. K. Srinivasa Rao	Jr. Clerk (joined on 20-8-1987)
23.	Sh. T. Pardha Sarathi	Jr. Clerk (Joined on 26-8-1987)
24.	Sh. P. V. V. S. S. R. L. Parthi Raj	Jr. Clerk (joined on 26-8-1987)

25.	Sh.G.Raj Reddy	Jr.Clerk (joined on 27-8-1987)
26.	Smt.Padma Saroja	Jr.Clerk (joined on 26-8-1987)
27.	Smt.B.Shashi Vardhini	Jr.Clerk (Joined on 27-8-1987)
28.	Sh.P.Sesha Sai Kumar	Jr.Clerk (Joined on 31-8-1987)
29.	Sh.S.Bala Kamesh	Jr.Clerk (Joined on 16-11-1987)

Hostel Services

1.	Sh.S.Papaiah	Museum Asstt. (Joined on 18-3-1987)
2.	Sh.Sham Bahadur	Catering-in-Charge
3.	Sh.P.Neelakantam	Sr.Clerk
4.	Smt.D.Sangamma	Cook (SSG II)
5.	Sh.B.Subba Rao	Asst.Cook (SSG.I)
6.	Smt.B.Santhamma	SSG.I
7.	Smt.K.Kalavathi	SSG.I
8.	Smt.S.Shakuntala	SSG.I
9.	Sh.G.V.Bikshapathi	SSG.I
10.	Sh.Phool Kumar	SSG.I
11.	Sh.A.Yadaiah	SSG.I
12.	Sh.M.Krishnaiah	SSG.I
13.	Sh.G.Dasaradha	SSG.I
14.	Sh.C.Anjaiah	SSG.I
15.	Smt.T.Jangamma	SSG.I

Other Supporting Services

1.	Sh.B.H.Satyanarayana	Security Officer-cum-Caretaker (Joined in 1-4-1987)
2.	Sh.L.Ramesh	Electrician-cum-Plumber
3.	Sh.K.Sattiah	Driver
4.	Sh.S.K.Peera	Driver
5.	Sh.Mohd.Naseer Khan	Driver
6.	Sh.G.Muthyalu	Driver
7.	Sh.U.Venkataratnam	Driver
8.	Sh.N.Ashok	Driver
9.	Smt.G.Mari Bai	SSG.II
10.	Sh.B.Narayana	SSG.II
11.	Sh.B.H.Dharma Raj	SSG.II
12.	Sh.P.Bal Raj	SSG.I
13.	Sh.Khalid	SSG.I
14.	Sh.M.Padmaiah	SSG.I
15.	Sh.M.Yadaiah	SSG.I
16.	Sh.K.Satyanarayana	SSG.I
17.	Sh.G.Pentaiah	SSG.I
18.	Sh.S.Jangaiah	SSG.I
19.	Smt.E.Bharatamma	SSG.I
20.	Sh.M.Narsing Rao	SSG.I
21.	Sh.M.Ashok	SSG.I

22.	Sh.C.Bikshapathi	SSG.I
23.	Sh.R.Sathaiah	SSG.I
24.	Sh.J.Chandriah	SSG.I
25.	Smt.N.Sukunamma	SSG.I
26.	Sh.M.Sham Rao	SSG.I
27.	Smt.M.Yadamma	SSG.I
28.	Sh.P.Iswari	SSG.I
29.	Sh.D.Papulu	SSG.I

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